

JNVST Class 6 Question Bank Booklet (Tamil Edition)

Real practice questions from the VedicNeev question bank, with full worked solutions.

Total Questions: 360 Easy: 133 Medium: 119 Hard: 108

Mental Ability (126 questions)

Q1. Find the missing number in the pattern: 1, 4, 9, 16, 25, ?

EASY

A. 36 (Correct)

B. 25

C. 35

D. 30

Explanation: These are consecutive squares starting from 1^2 ($1^2, 2^2, 3^2, 4^2, 5^2$); the next is $6^2 = 36$.

Q2. Find the missing number in the pattern: 4, 9, 16, 25, 36, ?

EASY

A. 36

B. 48

C. 42

D. 49 (Correct)

Explanation: These are consecutive squares starting from 2^2 ($2^2, 3^2, 4^2, 5^2, 6^2$); the next is $7^2 = 49$.

Q3. Find the missing number in the pattern: 9, 16, 25, 36, 49, ?

EASY

A. 64 (Correct)

B. 49

C. 63

D. 56

Explanation: These are consecutive squares starting from 3^2 ($3^2, 4^2, 5^2, 6^2, 7^2$); the next is $8^2 = 64$.

Q4. Find the missing number in the pattern: 16, 25, 36, 49, 64, ?

EASY

A. 64

B. 80

C. 72

D. 81 (Correct)

Explanation: These are consecutive squares starting from 4^2 ($4^2, 5^2, 6^2, 7^2, 8^2$); the next is $9^2 = 81$.

Q5. Find the missing number in the pattern: 25, 36, 49, 64, 81, ?

EASY

A. 100 (Correct)

B. 81

C. 99

D. 90

Explanation: These are consecutive squares starting from 5^2 ($5^2, 6^2, 7^2, 8^2, 9^2$); the next is $10^2 = 100$.

Q6. Find the missing number in the pattern: 1, 8, 27, 64, 125, ?

EASY

A. 216 (Correct)

B. 125

C. 215

D. 180

Explanation: These are consecutive cubes starting from 1^3 ($1^3, 2^3, 3^3, 4^3, 5^3$); the next is $6^3 = 216$.

Q7. Find the missing number in the pattern: 8, 27, 64, 125, 216, ?

EASY

A. 294

B. 343 (Correct)

C. 216

D. 342

Explanation: These are consecutive cubes starting from 2^3 ($2^3, 3^3, 4^3, 5^3, 6^3$); the next is $7^3 = 343$.

Q8. Find the missing number in the series: 3, 5, 7, 9, 11, ?

EASY

A. 12

B. 14

C. 9

D. 13 (Correct)

Explanation: The series increases by a fixed common difference of 2 each time (3 '5 '7 '9 '11 '13). Continue the pattern: $11 + 2 = 13$.

Q9. Find the missing number in the series: 5, 8, 11, 14, 17, ?

EASY

A. 20 (Correct)

B. 19

C. 21

D. 14

Explanation: The series increases by a fixed common difference of 3 each time (5 '8 '11 '14 '17 '20). Continue the pattern: $17 + 3 = 20$.

Q10. Find the missing number in the series: 7, 11, 15, 19, 23, ?

EASY

A. 19

B. 27 (Correct)

C. 26

D. 28

Explanation: The series increases by a fixed common difference of 4 each time (7 '11 '15 '19 '23 '27). Continue the pattern: $23 + 4 = 27$.

Q11. Find the missing number in the series: 2, 7, 12, 17, 22, ?

EASY

A. 17

B. 27 (Correct)

C. 26

D. 28

Explanation: The series increases by a fixed common difference of 5 each time (2 '7 '12 '17 '22 '27). Continue the pattern: $22 + 5 = 27$.

Q12. Find the missing number in the series: 6, 12, 18, 24, 30, ?

EASY

A. 36 (Correct)

B. 35

C. 37

D. 24

Explanation: The series increases by a fixed common difference of 6 each time (6 '12 '18 '24 '30 '36). Continue the pattern:
 $30 + 6 = 36$.

Q13. Find the missing number in the series: 4, 11, 18, 25, 32, ?

EASY

A. 25

B. 39 (Correct)

C. 38

D. 40

Explanation: The series increases by a fixed common difference of 7 each time (4 '11 '18 '25 '32 '39). Continue the pattern:
 $32 + 7 = 39$.

Q14. Find the missing number in the series: 9, 17, 25, 33, 41, ?

EASY

A. 48

B. 50

C. 33

D. 49 (Correct)

Explanation: The series increases by a fixed common difference of 8 each time (9 '17 '25 '33 '41 '49). Continue the pattern:
 $41 + 8 = 49$.

Q15. Find the missing number in the series: 3, 12, 21, 30, 39, ?

EASY

A. 48 (Correct)

B. 47

C. 49

D. 30

Explanation: The series increases by a fixed common difference of 9 each time (3 '12 '21 '30 '39 '48). Continue the pattern:
 $39 + 9 = 48$.

Q16. Find the missing number in the series: 8, 19, 30, 41, 52, ?

EASY

A. 41

B. 63 (Correct)

C. 62

D. 64

Explanation: The series increases by a fixed common difference of 11 each time (8 '19 '30 '41 '52 '63). Continue the pattern:
 $52 + 11 = 63$.

Q17. Find the missing number in the series: 5, 17, 29, 41, 53, ?

EASY

A. 64

B. 66

C. 41

D. 65 (Correct)

Explanation: The series increases by a fixed common difference of 12 each time (5 '17 '29 '41 '53 '65). Continue the pattern:
 $53 + 12 = 65$.

Q18. Find the missing number: 2, 4, 8, 16, ?, 64

EASY

- A. 24
- B. 32 (Correct)**
- C. 48
- D. 36

Explanation: Each number is double the previous one: $2 \times 2 = 4$, $4 \times 2 = 8$, $8 \times 2 = 16$, $16 \times 2 = 32$.

Q19. Find the missing letter: A, C, E, G, ?

EASY

- A. H
- B. I (Correct)**
- C. J
- D. F

Explanation: The series skips one letter each time (+2): ACEGI.

Q20. Find the odd one out: Carrot, Apple, Mango, Banana

EASY

- A. Banana
- B. Carrot (Correct)**
- C. Apple
- D. Mango

Explanation: Apple, Mango, Banana are all fruits; Carrot is not.

Q21. Find the odd one out: Sphere, Triangle, Square, Circle

EASY

- A. Sphere (Correct)**
- B. Triangle
- C. Square
- D. Circle

Explanation: Triangle, Square, Circle are all flat (2D) shapes; Sphere is not.

Q22. Find the odd one out: January, Monday, Wednesday, Friday

EASY

- A. Monday
- B. Wednesday
- C. Friday
- D. January (Correct)**

Explanation: Monday, Wednesday, Friday are all days of the week; January is not.

Q23. Find the odd one out: 9, 2, 4, 8

EASY

- A. 4
- B. 8
- C. 9 (Correct)**
- D. 2

Explanation: 2, 4, 8 are all even numbers; 9 is not.

Q24. Find the odd one out: Sparrow, Cow, Horse, Goat

EASY

- A. Goat
- B. Sparrow (Correct)**
- C. Cow
- D. Horse

Explanation: Cow, Horse, Goat are all land mammals; Sparrow is not.

Q25. Find the odd one out: Cricket bat, Guitar, Violin, Flute

EASY

- A. Cricket bat (Correct)**
- B. Guitar
- C. Violin
- D. Flute

Explanation: Guitar, Violin, Flute are all musical instruments; Cricket bat is not.

Q26. Find the odd one out: Asia, India, Japan, Brazil

EASY

- A. India
- B. Japan
- C. Brazil
- D. Asia (Correct)**

Explanation: India, Japan, Brazil are all countries; Asia is not.

Q27. Find the odd one out: Heart, Eye, Ear, Nose

EASY

- A. Ear
- B. Nose
- C. Heart (Correct)**
- D. Eye

Explanation: Eye, Ear, Nose are all sense organs; Heart is not.

Q28. Find the odd one out: Aeroplane, Car, Bus, Train

EASY

- A. Train
- B. Aeroplane (Correct)**
- C. Car
- D. Bus

Explanation: Car, Bus, Train are all land vehicles; Aeroplane is not.

Q29. Find the odd one out: 9, 3, 5, 7

EASY

- A. 7
- B. 9 (Correct)**
- C. 3
- D. 5

Explanation: 3, 5, 7 are all prime numbers; 9 is not.

Q30. இவற்றில் எது மற்றவற்றுடன் பொருந்தவில்லை: முக்கோணம், சதுரம், வட்டம், கோளம்?

EASY

- A. முக்கோணம்
- B. சதுரம்
- C. வட்டம்
- D. கோளம் (Correct)**

Explanation: முக்கோணம், சதுரம், வட்டம் ஆகியவை தட்டையான (2D) வடிவங்கள்; கோளம் ஒரு திண்மையான (3D) வடிவம்.

Q31. Which one does not belong with the others: Apple, Banana, Carrot, Mango?

EASY

- A. Apple / 8G,
- B. Banana / G2>
- C. Carrot / ✗(Correct)**
- D. Mango / .

Explanation: Carrot is a vegetable; the rest are fruits.

Q32. The reference figure is rotated 90° clockwise. Which option (A-D) shows the correct result?

EASY

- A. A
- B. B (Correct)**
- C. C
- D. D

Explanation: Rotating the reference by 90° clockwise gives the figure shown in option B.

Q33. The reference figure is rotated 180° clockwise. Which option (A-D) shows the correct result?

EASY

- A. A
- B. B
- C. C (Correct)**
- D. D

Explanation: Rotating the reference by 180° clockwise gives the figure shown in option C.

Q34. The reference figure is rotated 270° clockwise. Which option (A-D) shows the correct result?

EASY

- A. A
- B. B
- C. C
- D. D (Correct)**

Explanation: Rotating the reference by 270° clockwise gives the figure shown in option D.

Q35. The reference figure is rotated 90° clockwise. Which option (A-D) shows the correct result?

EASY

- A. A
- B. B (Correct)**
- C. C
- D. D

Explanation: Rotating the reference by 90° clockwise gives the figure shown in option B.

Q36. The reference figure is rotated 180° clockwise. Which option (A-D) shows the correct result?

EASY

- A. A
- B. B
- C. C (Correct)**
- D. D

Explanation: Rotating the reference by 180° clockwise gives the figure shown in option C.

Q37. The reference figure is rotated 270° clockwise. Which option (A-D) shows the correct result?

EASY

- A. A
- B. B
- C. C
- D. D (Correct)**

Explanation: Rotating the reference by 270° clockwise gives the figure shown in option D.

Q38. The reference figure is rotated 90° clockwise. Which option (A-D) shows the correct result?

EASY

- A. A
- B. B (Correct)**
- C. C
- D. D

Explanation: Rotating the reference by 90° clockwise gives the figure shown in option B.

Q39. The reference figure is rotated 180° clockwise. Which option (A-D) shows the correct result?

EASY

- A. A
- B. B
- C. C (Correct)**
- D. D

Explanation: Rotating the reference by 180° clockwise gives the figure shown in option C.

Q40. The reference figure is rotated 90° clockwise. Which option (A-D) shows the correct result?

EASY

- A. A
- B. B (Correct)**
- C. C
- D. D

Explanation: Rotating the reference by 90° clockwise gives the figure shown in option B.

Q41. The reference figure is rotated 270° clockwise. Which option (A-D) shows the correct result?

EASY

- A. A
- B. B
- C. C
- D. D (Correct)**

Explanation: Rotating the reference by 270° clockwise gives the figure shown in option D.

Q42. The figure rotates by 90° clockwise at each step in the series. Which option (A-D) comes next?

EASY

- A. A (Correct)**
- B. B
- C. C
- D. D

Explanation: Each term adds 90° to the last (clockwise); the next term is $270^\circ + 90^\circ = 0^\circ \pmod{360^\circ}$, shown in option A.

Q43. The figure rotates by 90° clockwise at each step in the series. Which option (A-D) comes next?

EASY

- A. A
- B. B (Correct)**
- C. C
- D. D

Explanation: Each term adds 90° to the last (clockwise); the next term is $0^\circ + 90^\circ = 90^\circ \pmod{360^\circ}$, shown in option B.

Q44. The figure rotates by 90° clockwise at each step in the series. Which option (A-D) comes next?

EASY

- A. A
- B. B
- C. C (Correct)**
- D. D

Explanation: Each term adds 90° to the last (clockwise); the next term is $270^\circ + 90^\circ = 0^\circ \pmod{360^\circ}$, shown in option C.

Q45. The figure rotates by 90° clockwise at each step in the series. Which option (A-D) comes next?

EASY

- A. A
- B. B
- C. C
- D. D (Correct)**

Explanation: Each term adds 90° to the last (clockwise); the next term is $90^\circ + 90^\circ = 180^\circ \pmod{360^\circ}$, shown in option D.

Q46. The figure rotates by 90° clockwise at each step in the series. Which option (A-D) comes next?

EASY

- A. A (Correct)**
- B. B
- C. C
- D. D

Explanation: Each term adds 90° to the last (clockwise); the next term is $270^\circ + 90^\circ = 0^\circ \pmod{360^\circ}$, shown in option A.

Q47. The figure rotates by 90° clockwise at each step in the series. Which option (A-D) comes next?

EASY

- A. A
- B. B (Correct)**
- C. C
- D. D

Explanation: Each term adds 90° to the last (clockwise); the next term is $0^\circ + 90^\circ = 90^\circ \pmod{360^\circ}$, shown in option B.

Q48. The figure rotates by 90° clockwise at each step in the series. Which option (A-D) comes next?

EASY

- A. A
- B. B
- C. C (Correct)**
- D. D

Explanation: Each term adds 90° to the last (clockwise); the next term is $270^\circ + 90^\circ = 0^\circ \pmod{360^\circ}$, shown in option C.

Q49. The figure rotates by 90° clockwise at each step in the series. Which option (A-D) comes next?

EASY

- A. A
- B. B
- C. C
- D. D (Correct)**

Explanation: Each term adds 90° to the last (clockwise); the next term is $0^\circ + 90^\circ = 90^\circ \pmod{360^\circ}$, shown in option D.

Q50. The figure rotates by 180° clockwise at each step in the series. Which option (A-D) comes next?

EASY

- A. A (Correct)**
- B. B
- C. C
- D. D

Explanation: Each term adds 180° to the last (clockwise); the next term is $180^\circ + 180^\circ = 0^\circ \pmod{360^\circ}$, shown in option A.

Q51. If CAT = 24 (using each letter's position in the alphabet, A=1, B=2, ...), then DOG = ?

MEDIUM

- A. 55
- B. 6
- C. 26 (Correct)**
- D. 27

Explanation: Each letter is coded by its position in the alphabet (A=1 ... Z=26) and the code is their sum. For DOG: D=4, O=15, G=7 'sum = 26.

Q52. If DOG = 26 (using each letter's position in the alphabet, A=1, B=2, ...), then CAT = ?

MEDIUM

- A. 24 (Correct)**
- B. 25
- C. 57
- D. 17

Explanation: Each letter is coded by its position in the alphabet (A=1 ... Z=26) and the code is their sum. For CAT: C=3, A=1, T=20 'sum = 24.

Q53. If SUN = 54 (using each letter's position in the alphabet, A=1, B=2, ...), then MAP = ?

MEDIUM

- A. 51
- B. 16
- C. 30 (Correct)**
- D. 31

Explanation: Each letter is coded by its position in the alphabet (A=1 ... Z=26) and the code is their sum. For MAP: M=13, A=1, P=16 'sum = 30.

Q54. If BAT = 23 (using each letter's position in the alphabet, A=1, B=2, ...), then SUN = ?

MEDIUM

- A. 27
- B. 34
- C. 54 (Correct)**
- D. 55

Explanation: Each letter is coded by its position in the alphabet (A=1 ... Z=26) and the code is their sum. For SUN: S=19, U=21, N=14 'sum = 54.

Q55. If MAP = 30 (using each letter's position in the alphabet, A=1, B=2, ...), then BAT = ?

MEDIUM

- A. 7
- B. 23 (Correct)**
- C. 24
- D. 58

Explanation: Each letter is coded by its position in the alphabet (A=1 ... Z=26) and the code is their sum. For BAT: B=2, A=1, T=20 'sum = 23.

Q56. In each row below, the same rule turns the first two numbers into the third. Find the missing number:

Row 1: 2, 3, 5

Row 2: 4, 5, 19

Row 3: 6, 7, ?

MEDIUM

- A. 42
- B. 12
- C. 43
- D. 41 (Correct)**

Explanation: The rule is: third number = (first × second) + (first – second). Checking row 1: $2 \times 3 + (2 - 3) = 5$. Applying it to row 3: $6 \times 7 + (6 - 7) = 41$.

Q57. In each row below, the same rule turns the first two numbers into the third. Find the missing number:

Row 1: 3, 4, 11

Row 2: 5, 6, 29

Row 3: 7, 8, ?

MEDIUM

- A. 57
- B. 55 (Correct)**
- C. 56
- D. 14

Explanation: The rule is: third number = (first × second) + (first – second). Checking row 1: $3 \times 4 + (3 - 4) = 11$. Applying it to row 3: $7 \times 8 + (7 - 8) = 55$.

Q58. In each row below, the same rule turns the first two numbers into the third. Find the missing number:

Row 1: 2, 5, 7

Row 2: 3, 6, 15

Row 3: 4, 9, ?

MEDIUM

- A. 41
- B. 31 (Correct)**
- C. 36
- D. 8

Explanation: The rule is: third number = (first × second) + (first – second). Checking row 1: $2 \times 5 + (2 - 5) = 7$. Applying it to row 3: $4 \times 9 + (4 - 9) = 31$.

Q59. In each row below, the same rule turns the first two numbers into the third. Find the missing number:

Row 1: 1, 6, 1

Row 2: 2, 7, 9

Row 3: 3, 9, ?

MEDIUM

A. 27

B. 6

C. 33

D. 21 (Correct)

Explanation: The rule is: third number = (first × second) + (first – second). Checking row 1: $1 \times 6 + (1 - 6) = 1$. Applying it to row 3: $3 \times 9 + (3 - 9) = 21$.

Q60. In each row below, the same rule turns the first two numbers into the third. Find the missing number:

Row 1: 4, 2, 10

Row 2: 6, 3, 21

Row 3: 8, 5, ?

MEDIUM

A. 37

B. 43 (Correct)

C. 40

D. 16

Explanation: The rule is: third number = (first × second) + (first – second). Checking row 1: $4 \times 2 + (4 - 2) = 10$. Applying it to row 3: $8 \times 5 + (8 - 5) = 43$.

Q61. வடிவத்தில் விடுபட்ட எண்ணைக் கண்டறியவும்: 1, 4, 9, 16, 25, ?

MEDIUM

A. 30

B. 36 (Correct)

C. 32

D. 49

Explanation: இவை முழு வர்க்கங்கள் ($1^2, 2^2, 3^2, 4^2, 5^2, 6^2$); அடுத்தது $6^2 = 36$.

Q62. 7 : 14 :: 9 : ?

MEDIUM

A. 17

B. 20

C. 18 (Correct)

D. 19

Explanation: The rule is: double the first number (714). Applying it to 9: double the 9 = 18.

Q63. 4 : 12 :: 6 : ?

MEDIUM

A. 17

B. 20

C. 18 (Correct)

D. 19

Explanation: The rule is: triple the first number (412). Applying it to 6: triple the 6 = 18.

Q64. 10 : 15 :: 12 : ?

MEDIUM

- A. 18
- B. 16
- C. 19
- D. 17 (Correct)**

Explanation: The rule is: add 5 to the first number (1015). Applying it to 12: add 5 to the 12 = 17.

Q65. 5 : 25 :: 7 : ?

MEDIUM

- A. 50
- B. 48
- C. 51
- D. 49 (Correct)**

Explanation: The rule is: square the first number (525). Applying it to 7: square the 7 = 49.

Q66. 3 : 11 :: 5 : ?

MEDIUM

- A. 21
- B. 19 (Correct)**
- C. 20
- D. 18

Explanation: The rule is: multiply the first number by 4 and subtract 1 (311). Applying it to 5: multiply the 5 by 4 and subtract 1 = 19.

Q67. Find the missing number in the pattern: 3, 9, 27, 81, ?

MEDIUM

- A. 162
- B. 202
- C. 243 (Correct)**
- D. 324

Explanation: Each number is multiplied by 3 to get the next: $81 \times 3 = 243$.

Q68. Find the missing number in the series: 10, 16, 14, 20, 18, ?

MEDIUM

- A. 24 (Correct)**
- B. 16
- C. 12
- D. 22

Explanation: The series alternates between adding 6 and subtracting 2 (+6, -2, +6, -2, ...). The last step was -2, so the next step adds 6: $18 + 6 = 24$.

Q69. Find the missing number in the series: 5, 13, 10, 18, 15, ?

MEDIUM

- A. 21
- B. 23 (Correct)**
- C. 12
- D. 7

Explanation: The series alternates between adding 8 and subtracting 3 (+8, -3, +8, -3, ...). The last step was -3, so the next step adds 8: $15 + 8 = 23$.

Q70. Find the missing number in the series: 20, 25, 16, 21, 12, ?

MEDIUM

- A. 3
- B. 7
- C. 19
- D. 17 (Correct)**

Explanation: The series alternates between adding 5 and subtracting 9 (+5, -9, +5, -9, ...). The last step was -9, so the next step adds 5: $12 + 5 = 17$.

Q71. Find the missing number in the series: 15, 22, 18, 25, 21, ?

MEDIUM

- A. 28 (Correct)**
- B. 17
- C. 14
- D. 27

Explanation: The series alternates between adding 7 and subtracting 4 (+7, -4, +7, -4, ...). The last step was -4, so the next step adds 7: $21 + 7 = 28$.

Q72. Find the missing number in the series: 8, 18, 15, 25, 22, ?

MEDIUM

- A. 32 (Correct)**
- B. 19
- C. 12
- D. 29

Explanation: The series alternates between adding 10 and subtracting 3 (+10, -3, +10, -3, ...). The last step was -3, so the next step adds 10: $22 + 10 = 32$.

Q73. Find the missing number in the series: 2, 4, 8, 16, 32, ?

MEDIUM

- A. 64 (Correct)**
- B. 48
- C. 96
- D. 32

Explanation: Each term is 2 times the one before it (a geometric series), so the next term is $32 \times 2 = 64$.

Q74. Find the missing number in the series: 3, 6, 12, 24, 48, ?

MEDIUM

- A. 96 (Correct)**
- B. 72
- C. 144
- D. 48

Explanation: Each term is 2 times the one before it (a geometric series), so the next term is $48 \times 2 = 96$.

Q75. Find the missing number in the series: 1, 3, 9, 27, 81, ?

MEDIUM

- A. 81
- B. 243 (Correct)**
- C. 108
- D. 324

Explanation: Each term is 3 times the one before it (a geometric series), so the next term is $81 \times 3 = 243$.

Q76. Find the missing number in the series: 4, 8, 16, 32, 64, ?

MEDIUM

A. 128 (Correct)

B. 96

C. 192

D. 64

Explanation: Each term is 2 times the one before it (a geometric series), so the next term is $64 \times 2 = 128$.

Q77. Find the missing number in the series: 2, 6, 18, 54, 162, ?

MEDIUM

A. 648

B. 162

C. 486 (Correct)

D. 216

Explanation: Each term is 3 times the one before it (a geometric series), so the next term is $162 \times 3 = 486$.

Q78. Find the missing number in the series: 1, 3, 7, 13, 21, ?

MEDIUM

A. 23

B. 31 (Correct)

C. 29

D. 32

Explanation: The differences between consecutive terms themselves increase by 2 each time (2, 4, 6, 8, ...), so the next difference is 10: $21 + 10 = 31$.

Q79. Find the missing number in the series: 2, 5, 10, 17, 26, ?

MEDIUM

A. 35

B. 38

C. 28

D. 37 (Correct)

Explanation: The differences between consecutive terms themselves increase by 2 each time (3, 5, 7, 9, ...), so the next difference is 11: $26 + 11 = 37$.

Q80. Find the missing number in the series: 0, 1, 5, 12, 22, ?

MEDIUM

A. 25

B. 35 (Correct)

C. 32

D. 36

Explanation: The differences between consecutive terms themselves increase by 3 each time (1, 4, 7, 10, ...), so the next difference is 13: $22 + 13 = 35$.

Q81. Find the missing number in the series: 3, 5, 10, 18, 29, ?

MEDIUM

A. 32

B. 43 (Correct)

C. 40

D. 44

Explanation: The differences between consecutive terms themselves increase by 3 each time (2, 5, 8, 11, ...), so the next difference is 14: $29 + 14 = 43$.

Q82. Find the missing number in the series: 1, 5, 11, 19, 29, ?

MEDIUM

- A. 39
- B. 42
- C. 31
- D. 41 (Correct)**

Explanation: The differences between consecutive terms themselves increase by 2 each time (4, 6, 8, 10, ...), so the next difference is 12: $29 + 12 = 41$.

Q83. Find the missing number: 5, 10, 20, 40, ?

MEDIUM

- A. 60
- B. 70
- C. 80 (Correct)**
- D. 45

Explanation: Each number doubles the previous one: $40 \times 2 = 80$.

Q84. Find the odd one out: 20, 12, 18, 24

MEDIUM

- A. 18
- B. 24
- C. 20 (Correct)**
- D. 12

Explanation: 12, 18, 24 are all multiples of 6; 20 is not.

Q85. Find the odd one out: Rectangle, Square, Equilateral triangle, Regular hexagon

MEDIUM

- A. Regular hexagon
- B. Rectangle (Correct)**
- C. Square
- D. Equilateral triangle

Explanation: Square, Equilateral triangle, Regular hexagon are all regular polygons (all sides and all angles equal); Rectangle is not.

Q86. Find the odd one out: Pluto, Mercury, Venus, Mars

MEDIUM

- A. Pluto (Correct)**
- B. Mercury
- C. Venus
- D. Mars

Explanation: Mercury, Venus, Mars are all planets of the solar system; Pluto is not.

Q87. Find the odd one out: Table, Run, Jump, Write

MEDIUM

- A. Run
- B. Jump
- C. Write
- D. Table (Correct)**

Explanation: Run, Jump, Write are all verbs (action words); Table is not.

Q88. Find the odd one out: Oxygen, Ice, Iron, Wood

MEDIUM

- A. Iron
- B. Wood
- C. Oxygen (Correct)**
- D. Ice

Explanation: Ice, Iron, Wood are all solids; Oxygen is not.

Q89. 3 : 10 :: 5 : ?

HARD

- A. 27
- B. 11
- C. 26 (Correct)**
- D. 25

Explanation: The rule is "square the first number, then add 1" ($3^2+1=10=10$). Applying it to 5: $5^2+1=25+1=26$.

Q90. 4 : 17 :: 6 : ?

HARD

- A. 36
- B. 38
- C. 13
- D. 37 (Correct)**

Explanation: The rule is "square the first number, then add 1" ($4^2+1=17=17$). Applying it to 6: $6^2+1=36+1=37$.

Q91. 2 : 5 :: 7 : ?

HARD

- A. 51
- B. 15
- C. 50 (Correct)**
- D. 49

Explanation: The rule is "square the first number, then add 1" ($2^2+1=5=5$). Applying it to 7: $7^2+1=49+1=50$.

Q92. 5 : 26 :: 3 : ?

HARD

- A. 11
- B. 7
- C. 10 (Correct)**
- D. 9

Explanation: The rule is "square the first number, then add 1" ($5^2+1=26=26$). Applying it to 3: $3^2+1=9+1=10$.

Q93. 6 : 37 :: 4 : ?

HARD

- A. 16
- B. 18
- C. 9
- D. 17 (Correct)**

Explanation: The rule is "square the first number, then add 1" ($6^2+1=37=37$). Applying it to 4: $4^2+1=16+1=17$.

Q94. In a code, CAT is written as ECV (each letter shifted forward by a fixed number of places in the alphabet). Using the same rule, if the code for DOG is coded the same way and each letter's alphabet position (A=1...Z=26) is summed, what is that total?

HARD

A. 32 (Correct)

B. 29

C. 35

D. 20

Explanation: CATECV shows a forward shift of 2 letters (e.g. CE). Applying the same +2 shift to DOG gives FQI, whose letter-position sum is 32.

Q95. In a code, DOG is written as GRJ (each letter shifted forward by a fixed number of places in the alphabet). Using the same rule, if the code for CAT is coded the same way and each letter's alphabet position (A=1...Z=26) is summed, what is that total?

HARD

A. 30

B. 36

C. 67

D. 33 (Correct)

Explanation: DOGGRJ shows a forward shift of 3 letters (e.g. DG). Applying the same +3 shift to CAT gives FDW, whose letter-position sum is 33.

Q96. In a code, SUN is written as WYR (each letter shifted forward by a fixed number of places in the alphabet). Using the same rule, if the code for BAT is coded the same way and each letter's alphabet position (A=1...Z=26) is summed, what is that total?

HARD

A. 63

B. 35 (Correct)

C. 32

D. 38

Explanation: SUNWYR shows a forward shift of 4 letters (e.g. SW). Applying the same +4 shift to BAT gives FEX, whose letter-position sum is 35.

Q97. In a code, MAP is written as RFU (each letter shifted forward by a fixed number of places in the alphabet). Using the same rule, if the code for SUN is coded the same way and each letter's alphabet position (A=1...Z=26) is summed, what is that total?

HARD

A. 66

B. 46

C. 39

D. 69 (Correct)

Explanation: MAPRFU shows a forward shift of 5 letters (e.g. MR). Applying the same +5 shift to SUN gives XZS, whose letter-position sum is 69.

Q98. In a code, BAT is written as CBU (each letter shifted forward by a fixed number of places in the alphabet). Using the same rule, if the code for MAP is coded the same way and each letter's alphabet position (A=1...Z=26) is summed, what is that total?

HARD

A. 30

B. 36

C. 53

D. 33 (Correct)

Explanation: BATCBU shows a forward shift of 1 letters (e.g. BC). Applying the same +1 shift to MAP gives NBQ, whose letter-position sum is 33.

Q99. In this 3×3 grid, every row adds up to the same total. Find the missing number: 4, 9, 2 | 3, 5, 7 | 8, 1, ?

HARD

- A. 8
- B. 9
- C. 6 (Correct)
- D. 7

Explanation: Every row sums to 15 (row 1: $4+9+2=15$; row 2: $3+5+7=15$). So row 3's missing number is $15 - 8 - 1 = 6$.

Q100. In this 3×3 grid, every row adds up to the same total. Find the missing number: 6, 1, 8 | 7, 5, 3 | 2, 9, ?

HARD

- A. 4 (Correct)
- B. 13
- C. 6
- D. 11

Explanation: Every row sums to 15 (row 1: $6+1+8=15$; row 2: $7+5+3=15$). So row 3's missing number is $15 - 2 - 9 = 4$.

Q101. In this 3×3 grid, every row adds up to the same total. Find the missing number: 2, 7, 6 | 9, 5, 1 | 4, 3, ?

HARD

- A. 8 (Correct)
- B. 11
- C. 10
- D. 7

Explanation: Every row sums to 15 (row 1: $2+7+6=15$; row 2: $9+5+1=15$). So row 3's missing number is $15 - 4 - 3 = 8$.

Q102. In this 3×3 grid, every row adds up to the same total. Find the missing number: 8, 3, 4 | 1, 5, 9 | 6, 7, ?

HARD

- A. 4
- B. 13
- C. 2 (Correct)
- D. 9

Explanation: Every row sums to 15 (row 1: $8+3+4=15$; row 2: $1+5+9=15$). So row 3's missing number is $15 - 6 - 7 = 2$.

Q103. In this 3×3 grid, every row adds up to the same total. Find the missing number: 1, 5, 9 | 8, 3, 4 | 6, 7, ?

HARD

- A. 4
- B. 13
- C. 2 (Correct)
- D. 9

Explanation: Every row sums to 15 (row 1: $1+5+9=15$; row 2: $8+3+4=15$). So row 3's missing number is $15 - 6 - 7 = 2$.

Q104. Find the missing number in the series: 1, 1, 2, 3, 5, ?

HARD

- A. 8 (Correct)
- B. 5
- C. 10
- D. 6

Explanation: Each term is the sum of the two terms before it ($1+1=2$, $1+2=3$, $2+3=5$), so the next term is $3+5=8$.

Q105. Find the missing number in the series: 2, 3, 5, 8, 13, ?

HARD

- A. 13
- B. 26
- C. 16
- D. 21 (Correct)**

Explanation: Each term is the sum of the two terms before it ($2+3=5$, $3+5=8$, $5+8=13$), so the next term is $8+13=21$.

Q106. Find the missing number in the series: 1, 4, 5, 9, 14, ?

HARD

- A. 18
- B. 23 (Correct)**
- C. 14
- D. 28

Explanation: Each term is the sum of the two terms before it ($1+4=5$, $4+5=9$, $5+9=14$), so the next term is $9+14=23$.

Q107. Find the missing number in the series: 3, 5, 8, 13, 21, ?

HARD

- A. 42
- B. 26
- C. 34 (Correct)**
- D. 21

Explanation: Each term is the sum of the two terms before it ($3+5=8$, $5+8=13$, $8+13=21$), so the next term is $13+21=34$.

Q108. Find the missing number in the series: 2, 2, 4, 6, 10, ?

HARD

- A. 16 (Correct)**
- B. 10
- C. 20
- D. 12

Explanation: Each term is the sum of the two terms before it ($2+2=4$, $2+4=6$, $4+6=10$), so the next term is $6+10=16$.

Q109. Find the missing number in the series: 1, 10, 3, 15, 5, ?

HARD

- A. 20 (Correct)**
- B. 7
- C. 17
- D. 10

Explanation: This is actually two interleaved series: the 1st/3rd/5th positions (1, 3, 5) rise by 2, and the 2nd/4th positions (10, 15) rise by 5. The missing 6th term continues the second series: $15 + 5 = 20$.

Q110. Find the missing number in the series: 3, 20, 6, 24, 9, ?

HARD

- A. 28 (Correct)**
- B. 12
- C. 27
- D. 13

Explanation: This is actually two interleaved series: the 1st/3rd/5th positions (3, 6, 9) rise by 3, and the 2nd/4th positions (20, 24) rise by 4. The missing 6th term continues the second series: $24 + 4 = 28$.

Q111. Find the missing number in the series: 2, 15, 6, 21, 10, ?

HARD

- A. 16
- B. 27 (Correct)**
- C. 14
- D. 25

Explanation: This is actually two interleaved series: the 1st/3rd/5th positions (2, 6, 10) rise by 4, and the 2nd/4th positions (15, 21) rise by 6. The missing 6th term continues the second series: $21 + 6 = 27$.

Q112. Find the missing number in the series: 5, 8, 7, 15, 9, ?

HARD

- A. 17
- B. 16
- C. 22 (Correct)**
- D. 11

Explanation: This is actually two interleaved series: the 1st/3rd/5th positions (5, 7, 9) rise by 2, and the 2nd/4th positions (8, 15) rise by 7. The missing 6th term continues the second series: $15 + 7 = 22$.

Q113. Find the missing number in the series: 1, 12, 6, 15, 11, ?

HARD

- A. 20
- B. 14
- C. 18 (Correct)**
- D. 16

Explanation: This is actually two interleaved series: the 1st/3rd/5th positions (1, 6, 11) rise by 5, and the 2nd/4th positions (12, 15) rise by 3. The missing 6th term continues the second series: $15 + 3 = 18$.

Q114. Find the missing number in the series: 1, 3, 7, 15, 31, ?

HARD

- A. 64
- B. 63 (Correct)**
- C. 62
- D. 61

Explanation: Each term is double the previous term plus 1 ($n \times 2 + 1$), so the next term is $31 \times 2 + 1 = 63$.

Q115. Find the missing number in the series: 2, 5, 11, 23, 47, ?

HARD

- A. 96
- B. 95 (Correct)**
- C. 94
- D. 93

Explanation: Each term is double the previous term plus 1 ($n \times 2 + 1$), so the next term is $47 \times 2 + 1 = 95$.

Q116. Find the missing number in the series: 1, 4, 10, 22, 46, ?

HARD

- A. 90
- B. 95
- C. 94 (Correct)**
- D. 92

Explanation: Each term is double the previous term plus 2 ($n \times 2 + 2$), so the next term is $46 \times 2 + 2 = 94$.

Q117. Find the missing number in the series: 3, 8, 18, 38, 78, ?

HARD

- A. 154
- B. 159
- C. 158 (Correct)**
- D. 156

Explanation: Each term is double the previous term plus 2 ($n \times 2n + 2$), so the next term is $78 \times 2 + 2 = 158$.

Q118. Find the missing number in the series: 2, 7, 17, 37, 77, ?

HARD

- A. 154
- B. 151
- C. 158
- D. 157 (Correct)**

Explanation: Each term is double the previous term plus 3 ($n \times 2n + 3$), so the next term is $77 \times 2 + 3 = 157$.

Q119. Find the odd one out: Bat, Sparrow, Eagle, Parrot

HARD

- A. Bat (Correct)**
- B. Sparrow
- C. Eagle
- D. Parrot

Explanation: Sparrow, Eagle, Parrot are all birds; Bat is not.

Q120. Find the odd one out: Whale, Shark, Salmon, Tuna

HARD

- A. Shark
- B. Salmon
- C. Tuna
- D. Whale (Correct)**

Explanation: Shark, Salmon, Tuna are all fish (breathe through gills); Whale is not.

Q121. Find the odd one out: Spider, Ant, Housefly, Butterfly

HARD

- A. Housefly
- B. Butterfly
- C. Spider (Correct)**
- D. Ant

Explanation: Ant, Housefly, Butterfly are all insects (six legs); Spider is not.

Q122. Find the odd one out: Tomato (botanically a fruit), Carrot (a root), Potato (a stem tuber), Spinach (a leaf)

HARD

- A. Spinach (a leaf)
- B. Tomato (botanically a fruit) (Correct)**
- C. Carrot (a root)
- D. Potato (a stem tuber)

Explanation: Carrot (a root), Potato (a stem tuber), Spinach (a leaf) are all vegetables — edible root, stem, or leaf parts of a plant; Tomato (botanically a fruit) is not.

Q123. Find the odd one out: Peanut, Almond, Cashew, Walnut

HARD

A. Peanut (Correct)

B. Almond

C. Cashew

D. Walnut

Explanation: Almond, Cashew, Walnut are all tree nuts; Peanut is not.

Q124. Find the odd one out: Penguin, Sparrow, Crow, Pigeon

HARD

A. Sparrow

B. Crow

C. Pigeon

D. Penguin (Correct)

Explanation: Sparrow, Crow, Pigeon are all birds that can fly; Penguin is not.

Q125. Find the odd one out: Diamond, Oxygen, Gold, Carbon

HARD

A. Gold

B. Carbon

C. Diamond (Correct)

D. Oxygen

Explanation: Oxygen, Gold, Carbon are all chemical elements; Diamond is not.

Q126. Find the odd one out: 50, 16, 25, 49

HARD

A. 49

B. 50 (Correct)

C. 16

D. 25

Explanation: 16, 25, 49 are all perfect squares; 50 is not.

Arithmetic (126 questions)

Q1. Calculate quickly using the "all from 9, last from 10" shortcut: $100 - 63 = ?$

EASY

A. 36

B. 38

C. 47

D. 37 (Correct)

Explanation: Subtract every digit of 63 from 9 except the last, which is subtracted from 10: $100 - 63 = 37$.

Q2. Calculate quickly using the "all from 9, last from 10" shortcut: $100 - 38 = ?$

EASY

A. 63

B. 72

C. 62 (Correct)

D. 61

Explanation: Subtract every digit of 38 from 9 except the last, which is subtracted from 10: $100 - 38 = 62$.

Q3. Calculate quickly using vertically-and-crosswise multiplication: $12 \times 43 = ?$

EASY

- A. 516 (Correct)**
- B. 546
- C. 526
- D. 506

Explanation: Units: $2 \times 3 = 6$. Cross-multiply and add: $1 \times 3 + 2 \times 4 = 11$. Leading: $1 \times 4 = 4$. Combine with carries: $12 \times 43 = 516$.

Q4. Calculate quickly using vertically-and-crosswise multiplication: $21 \times 14 = ?$

EASY

- A. 304
- B. 284
- C. 294 (Correct)**
- D. 264

Explanation: Units: $1 \times 4 = 4$. Cross-multiply and add: $2 \times 4 + 1 \times 1 = 9$. Leading: $2 \times 1 = 2$. Combine with carries: $21 \times 14 = 294$.

Q5. விரைவாகக் கணக்கிடுங்கள்: $72 \times 11 = ?$

EASY

- A. 772
- B. 792 (Correct)**
- C. 702
- D. 812

Explanation: $\times 11$ க்கான சாண்ட்விச் விதி: 7_2 நடு இலக்கம் $7+2=9$ '792.

Q6. Calculate quickly using the Nikhilam near-base shortcut: $98 \times 97 = ?$

EASY

- A. 9500
- B. 10106
- C. 9506 (Correct)**
- D. 9494

Explanation: Both numbers are close to 100, with deviations -2 and -3. Cross-add one number and the other's deviation ($98-3 = 95$), multiply by the base, then add the product of the deviations ($-2 \times -3 = 6$): $100 \times 95 + 6 = 9506$.

Q7. Calculate quickly using the Nikhilam near-base shortcut: $99 \times 96 = ?$

EASY

- A. 9504 (Correct)**
- B. 9496
- C. 9500
- D. 10304

Explanation: Both numbers are close to 100, with deviations -1 and -4. Cross-add one number and the other's deviation ($99-4 = 95$), multiply by the base, then add the product of the deviations ($-1 \times -4 = 4$): $100 \times 95 + 4 = 9504$.

Q8. Calculate quickly using the "square ending in 5" shortcut: $15^2 = ?$

EASY

- A. 125
- B. 25
- C. 220
- D. 225 (Correct)**

Explanation: For a number $10a+5$ (here $a=1$), the square is $a \times (a+1)$ followed by 25: $1 \times 2 = 2$, so $15^2 = 225$.

Q9. Calculate quickly using the "square ending in 5" shortcut: $25^2 = ?$

EASY

- A. 425
- B. 225
- C. 620
- D. 625 (Correct)**

Explanation: For a number $10a+5$ (here $a=2$), the square is $a \times (a+1)$ followed by 25: $2 \times 3 = 6$, so $25^2 = 625$.

Q10. Calculate quickly using the $\times 11$ shortcut: $23 \times 11 = ?$

EASY

- A. 34
- B. 230
- C. 276
- D. 253 (Correct)**

Explanation: Sandwich the digit sum of 23 between its own digits (carrying if the sum reaches 10 or more): $23 \times 11 = 253$.

Q11. Calculate quickly using the $\times 11$ shortcut: $41 \times 11 = ?$

EASY

- A. 492
- B. 451 (Correct)**
- C. 52
- D. 410

Explanation: Sandwich the digit sum of 41 between its own digits (carrying if the sum reaches 10 or more): $41 \times 11 = 451$.

Q12. Calculate quickly: $45 \times 11 = ?$

EASY

- A. 485
- B. 495 (Correct)**
- C. 450
- D. 545

Explanation: Sandwich rule for $\times 11$: 4_5 with the middle digit $4+5=9$ '495.

Q13. What is the HCF (Highest Common Factor) of 12 and 18?

EASY

- A. 6 (Correct)**
- B. 8
- C. 12
- D. 4

Explanation: The factors of 12 are {1, 2, 3, 4, 6, 12} and of 18 are {1, 2, 3, 6, 9, 18}. The largest number common to both lists is 6.

Q14. What is the HCF (Highest Common Factor) of 24 and 36?

EASY

- A. 14
- B. 12 (Correct)**
- C. 24
- D. 10

Explanation: The factors of 24 are {1, 2, 3, 4, 6, 8, 12, 24} and of 36 are {1, 2, 3, 4, 6, 9, 12, 18, 36}. The largest number common to both lists is 12.

Q15. What is the HCF (Highest Common Factor) of 15 and 25?

EASY

- A. 7
- B. 15
- C. 5 (Correct)**
- D. 3

Explanation: The factors of 15 are {1, 3, 5, 15} and of 25 are {1, 5, 25}. The largest number common to both lists is 5.

Q16. What is the HCF (Highest Common Factor) of 8 and 20?

EASY

- A. 6
- B. 8
- C. 2
- D. 4 (Correct)**

Explanation: The factors of 8 are {1, 2, 4, 8} and of 20 are {1, 2, 4, 5, 10, 20}. The largest number common to both lists is 4.

Q17. What is the HCF (Highest Common Factor) of 14 and 21?

EASY

- A. 7 (Correct)**
- B. 9
- C. 14
- D. 5

Explanation: The factors of 14 are {1, 2, 7, 14} and of 21 are {1, 3, 7, 21}. The largest number common to both lists is 7.

Q18. What is the HCF (Highest Common Factor) of 16 and 40?

EASY

- A. 10
- B. 8 (Correct)**
- C. 16
- D. 6

Explanation: The factors of 16 are {1, 2, 4, 8, 16} and of 40 are {1, 2, 4, 5, 8, 10, 20, 40}. The largest number common to both lists is 8.

Q19. What is the HCF (Highest Common Factor) of 18 and 30?

EASY

- A. 8
- B. 18
- C. 6 (Correct)**
- D. 4

Explanation: The factors of 18 are {1, 2, 3, 6, 9, 18} and of 30 are {1, 2, 3, 5, 6, 10, 15, 30}. The largest number common to both lists is 6.

Q20. What is the HCF (Highest Common Factor) of 27 and 45?

EASY

- A. 11
- B. 27
- C. 7
- D. 9 (Correct)**

Explanation: The factors of 27 are {1, 3, 9, 27} and of 45 are {1, 3, 5, 9, 15, 45}. The largest number common to both lists is 9.

Q21. What is the HCF (Highest Common Factor) of 10 and 15?

EASY

A. 5 (Correct)

- B. 7
- C. 10
- D. 3

Explanation: The factors of 10 are {1, 2, 5, 10} and of 15 are {1, 3, 5, 15}. The largest number common to both lists is 5.

Q22. What is the HCF (Highest Common Factor) of 22 and 33?

EASY

B. 11 (Correct)

- A. 13
- C. 22
- D. 9

Explanation: The factors of 22 are {1, 2, 11, 22} and of 33 are {1, 3, 11, 33}. The largest number common to both lists is 11.

Q23. Calculate: $\frac{2}{7} + \frac{3}{7} = ?$

EASY

A. $\frac{5}{7}$ (Correct)

- B. $\frac{5}{14}$
- C. $-\frac{1}{7}$
- D. $\frac{6}{7}$

Explanation: The denominators are already the same, so just add the numerators: $2 + 3 = 5$, giving $\frac{5}{7}$, which simplifies to $\frac{5}{7}$.

Q24. Calculate: $\frac{5}{9} - \frac{2}{9} = ?$

EASY

B. $\frac{1}{3}$ (Correct)

- A. $\frac{7}{18}$
- C. $\frac{7}{9}$
- D. $\frac{2}{3}$

Explanation: The denominators are already the same, so just subtract the numerators: $5 - 2 = 3$, giving $\frac{3}{9}$, which simplifies to $\frac{1}{3}$.

Q25. Calculate: $\frac{1}{5} + \frac{2}{5} = ?$

EASY

C. $\frac{3}{5}$ (Correct)

- A. $\frac{3}{10}$
- B. $-\frac{1}{5}$
- D. $\frac{4}{5}$

Explanation: The denominators are already the same, so just add the numerators: $1 + 2 = 3$, giving $\frac{3}{5}$, which simplifies to $\frac{3}{5}$.

Q26. Calculate: $\frac{7}{10} - \frac{3}{10} = ?$

EASY

D. $\frac{2}{5}$ (Correct)

- A. $\frac{1}{2}$
- B. $\frac{1}{1}$
- C. $\frac{3}{5}$

Explanation: The denominators are already the same, so just subtract the numerators: $7 - 3 = 4$, giving $\frac{4}{10}$, which simplifies to $\frac{2}{5}$.

Q27. Calculate: $\frac{3}{8} + \frac{2}{8} = ?$

EASY

A. $\frac{5}{8}$ (Correct)

B. $\frac{5}{16}$

C. $\frac{1}{8}$

D. $\frac{3}{4}$

Explanation: The denominators are already the same, so just add the numerators: $3 + 2 = 5$, giving $\frac{5}{8}$, which simplifies to $\frac{5}{8}$.

Q28. Calculate: $\frac{4}{11} + \frac{1}{11} = ?$

EASY

A. $\frac{5}{22}$

B. $\frac{5}{11}$ (Correct)

C. $\frac{3}{11}$

D. $\frac{6}{11}$

Explanation: The denominators are already the same, so just add the numerators: $4 + 1 = 5$, giving $\frac{5}{11}$, which simplifies to $\frac{5}{11}$.

Q29. Calculate: $\frac{6}{13} - \frac{2}{13} = ?$

EASY

A. $\frac{8}{13}$

B. $\frac{5}{13}$

C. $\frac{4}{13}$ (Correct)

D. $\frac{1}{13}$

Explanation: The denominators are already the same, so just subtract the numerators: $6 - 2 = 4$, giving $\frac{4}{13}$, which simplifies to $\frac{4}{13}$.

Q30. Calculate: $\frac{2}{9} + \frac{5}{9} = ?$

EASY

A. $\frac{7}{18}$

B. $-\frac{1}{3}$

C. $\frac{8}{9}$

D. $\frac{7}{9}$ (Correct)

Explanation: The denominators are already the same, so just add the numerators: $2 + 5 = 7$, giving $\frac{7}{9}$, which simplifies to $\frac{7}{9}$.

Q31. Calculate: $\frac{8}{15} - \frac{3}{15} = ?$

EASY

A. $\frac{1}{3}$ (Correct)

B. $\frac{11}{30}$

C. $\frac{11}{15}$

D. $\frac{2}{3}$

Explanation: The denominators are already the same, so just subtract the numerators: $8 - 3 = 5$, giving $\frac{5}{15}$, which simplifies to $\frac{1}{3}$.

Q32. Calculate: $\frac{1}{6} + \frac{2}{6} = ?$

EASY

A. $\frac{1}{4}$

B. $\frac{1}{2}$ (Correct)

C. $-\frac{1}{6}$

D. $\frac{1}{1}$

Explanation: The denominators are already the same, so just add the numerators: $1 + 2 = 3$, giving $\frac{3}{6}$, which simplifies to $\frac{1}{2}$.

Q33. A shopkeeper buys an item for ₹100 and sells it for ₹120. What is the profit percent?

EASY

- A. 20% (Correct)**
- B. 16.67%
- C. 25%
- D. 26%

Explanation: Profit = $|\text{₹}120 - \text{₹}100| = \text{₹}20$. Profit% = $(20 \div 100) \times 100 = 20\%$.

Q34. A shopkeeper buys an item for ₹200 and sells it for ₹180. What is the loss percent?

EASY

- A. 11.11%
- B. 10% (Correct)**
- C. 20%
- D. 15%

Explanation: Loss = $|\text{₹}180 - \text{₹}200| = \text{₹}20$. Loss% = $(20 \div 200) \times 100 = 10\%$.

Q35. A shopkeeper buys an item for ₹250 and sells it for ₹300. What is the profit percent?

EASY

- A. 16.67%
- B. 50%
- C. 20% (Correct)**
- D. 25%

Explanation: Profit = $|\text{₹}300 - \text{₹}250| = \text{₹}50$. Profit% = $(50 \div 250) \times 100 = 20\%$.

Q36. A shopkeeper buys an item for ₹400 and sells it for ₹360. What is the loss percent?

EASY

- A. 11.11%
- B. 40%
- C. 15%
- D. 10% (Correct)**

Explanation: Loss = $|\text{₹}360 - \text{₹}400| = \text{₹}40$. Loss% = $(40 \div 400) \times 100 = 10\%$.

Q37. A shopkeeper buys an item for ₹150 and sells it for ₹165. What is the profit percent?

EASY

- A. 10% (Correct)**
- B. 9.09%
- C. 15%
- D. 16%

Explanation: Profit = $|\text{₹}165 - \text{₹}150| = \text{₹}15$. Profit% = $(15 \div 150) \times 100 = 10\%$.

Q38. A shopkeeper buys an item for ₹500 and sells it for ₹450. What is the loss percent?

EASY

- A. 11.11%
- B. 10% (Correct)**
- C. 50%
- D. 15%

Explanation: Loss = $|\text{₹}450 - \text{₹}500| = \text{₹}50$. Loss% = $(50 \div 500) \times 100 = 10\%$.

Q39. A shopkeeper buys an item for ₹80 and sells it for ₹100. What is the profit percent?

EASY

- A. 20%
- B. 30%
- C. 25% (Correct)**
- D. 31%

Explanation: Profit = $|\text{₹}100 - \text{₹}80| = \text{₹}20$. Profit% = $(20 \div 80) \times 100 = 25\%$.

Q40. A shopkeeper buys an item for ₹600 and sells it for ₹540. What is the loss percent?

EASY

- A. 11.11%
- B. 60%
- C. 15%
- D. 10% (Correct)**

Explanation: Loss = $|\text{₹}540 - \text{₹}600| = \text{₹}60$. Loss% = $(60 \div 600) \times 100 = 10\%$.

Q41. A shopkeeper buys an item for ₹120 and sells it for ₹150. What is the profit percent?

EASY

- A. 25% (Correct)**
- B. 20%
- C. 30%
- D. 31%

Explanation: Profit = $|\text{₹}150 - \text{₹}120| = \text{₹}30$. Profit% = $(30 \div 120) \times 100 = 25\%$.

Q42. A shopkeeper buys an item for ₹300 and sells it for ₹270. What is the loss percent?

EASY

- A. 11.11%
- B. 10% (Correct)**
- C. 30%
- D. 15%

Explanation: Loss = $|\text{₹}270 - \text{₹}300| = \text{₹}30$. Loss% = $(30 \div 300) \times 100 = 10\%$.

Q43. Find the perimeter of a rectangle with length 10 units and width 6 units.

EASY

- A. 32 units (Correct)**
- B. 60 units
- C. 16 units
- D. 36 units

Explanation: Perimeter = $2 \times (\text{length} + \text{width}) = 2 \times (10 + 6) = 2 \times 16 = 32$ units.

Q44. Find the perimeter of a square with length 8 units and width 8 units.

EASY

- A. 64 units
- B. 32 units (Correct)**
- C. 16 units
- D. 36 units

Explanation: Perimeter = $2 \times (\text{length} + \text{width}) = 2 \times (8 + 8) = 2 \times 16 = 32$ units.

Q45. Find the perimeter of a rectangle with length 15 units and width 9 units.

EASY

- A. 135 units
- B. 24 units
- C. 48 units (Correct)**
- D. 52 units

Explanation: Perimeter = $2 \times (\text{length} + \text{width}) = 2 \times (15 + 9) = 2 \times 24 = 48$ units.

Q46. Find the perimeter of a square with length 12 units and width 12 units.

EASY

- A. 144 units
- B. 24 units
- C. 52 units
- D. 48 units (Correct)**

Explanation: Perimeter = $2 \times (\text{length} + \text{width}) = 2 \times (12 + 12) = 2 \times 24 = 48$ units.

Q47. Find the perimeter of a rectangle with length 20 units and width 14 units.

EASY

- A. 68 units (Correct)**
- B. 280 units
- C. 34 units
- D. 72 units

Explanation: Perimeter = $2 \times (\text{length} + \text{width}) = 2 \times (20 + 14) = 2 \times 34 = 68$ units.

Q48. Find the perimeter of a square with length 7 units and width 7 units.

EASY

- A. 49 units
- B. 28 units (Correct)**
- C. 14 units
- D. 32 units

Explanation: Perimeter = $2 \times (\text{length} + \text{width}) = 2 \times (7 + 7) = 2 \times 14 = 28$ units.

Q49. Find the perimeter of a rectangle with length 18 units and width 11 units.

EASY

- A. 198 units
- B. 29 units
- C. 58 units (Correct)**
- D. 62 units

Explanation: Perimeter = $2 \times (\text{length} + \text{width}) = 2 \times (18 + 11) = 2 \times 29 = 58$ units.

Q50. Find the perimeter of a rectangle with length 25 units and width 16 units.

EASY

- A. 400 units
- B. 41 units
- C. 86 units
- D. 82 units (Correct)**

Explanation: Perimeter = $2 \times (\text{length} + \text{width}) = 2 \times (25 + 16) = 2 \times 41 = 82$ units.

Q51. Calculate quickly using the "all from 9, last from 10" shortcut: $1000 - 587 = ?$

MEDIUM

- A. 412
- B. 414
- C. 423
- D. 413 (Correct)**

Explanation: Subtract every digit of 587 from 9 except the last, which is subtracted from 10: $1000 - 587 = 413$.

Q52. Calculate quickly using the "all from 9, last from 10" shortcut: $1000 - 234 = ?$

MEDIUM

- A. 767
- B. 776
- C. 766 (Correct)**
- D. 765

Explanation: Subtract every digit of 234 from 9 except the last, which is subtracted from 10: $1000 - 234 = 766$.

Q53. Calculate quickly using the "all from 9, last from 10" shortcut: $1000 - 456 = ?$

MEDIUM

- A. 544 (Correct)**
- B. 543
- C. 545
- D. 554

Explanation: Subtract every digit of 456 from 9 except the last, which is subtracted from 10: $1000 - 456 = 544$.

Q54. Calculate quickly using vertically-and-crosswise multiplication: $34 \times 56 = ?$

MEDIUM

- A. 1904 (Correct)**
- B. 1944
- C. 1914
- D. 1894

Explanation: Units: $4 \times 6 = 24$. Cross-multiply and add: $3 \times 6 + 4 \times 5 = 38$. Leading: $3 \times 5 = 15$. Combine with carries: $34 \times 56 = 1904$.

Q55. Calculate quickly using vertically-and-crosswise multiplication: $47 \times 68 = ?$

MEDIUM

- A. 3196 (Correct)**
- B. 3216
- C. 3206
- D. 3186

Explanation: Units: $7 \times 8 = 56$. Cross-multiply and add: $4 \times 8 + 7 \times 6 = 74$. Leading: $4 \times 6 = 24$. Combine with carries: $47 \times 68 = 3196$.

Q56. Calculate quickly using vertically-and-crosswise multiplication: $63 \times 42 = ?$

MEDIUM

- A. 2656
- B. 2636
- C. 2646 (Correct)**
- D. 2666

Explanation: Units: $3 \times 2 = 6$. Cross-multiply and add: $6 \times 2 + 3 \times 4 = 24$. Leading: $6 \times 4 = 24$. Combine with carries: $63 \times 42 = 2646$.

Q57. Calculate quickly using the Nikhilam near-base shortcut: $103 \times 104 = ?$

MEDIUM

A. 10712 (Correct)

B. 10688

C. 10700

D. 9912

Explanation: Both numbers are close to 100, with deviations +3 and +4. Cross-add one number and the other's deviation ($103+4 = 107$), multiply by the base, then add the product of the deviations ($3 \times 4 = 12$): $100 \times 107 + 12 = 10712$.

Q58. Calculate quickly using the Nikhilam near-base shortcut: $92 \times 88 = ?$

MEDIUM

A. 8096 (Correct)

B. 7904

C. 8000

D. 10496

Explanation: Both numbers are close to 100, with deviations -8 and -12. Cross-add one number and the other's deviation ($92-12 = 80$), multiply by the base, then add the product of the deviations ($-8 \times -12 = 96$): $100 \times 80 + 96 = 8096$.

Q59. Calculate quickly using the Nikhilam near-base shortcut: $107 \times 102 = ?$

MEDIUM

A. 10900

B. 10514

C. 10914 (Correct)

D. 10886

Explanation: Both numbers are close to 100, with deviations +7 and +2. Cross-add one number and the other's deviation ($107+2 = 109$), multiply by the base, then add the product of the deviations ($7 \times 2 = 14$): $100 \times 109 + 14 = 10914$.

Q60. Calculate quickly using the "square ending in 5" shortcut: $65^2 = ?$

MEDIUM

A. 3625

B. 3025

C. 4220

D. 4225 (Correct)

Explanation: For a number $10a+5$ (here $a=6$), the square is $a \times (a+1)$ followed by 25: $6 \times 7 = 42$, so $65^2 = 4225$.

Q61. Calculate quickly using the "square ending in 5" shortcut: $75^2 = ?$

MEDIUM

A. 4925

B. 4225

C. 5620

D. 5625 (Correct)

Explanation: For a number $10a+5$ (here $a=7$), the square is $a \times (a+1)$ followed by 25: $7 \times 8 = 56$, so $75^2 = 5625$.

Q62. Calculate quickly using the "square ending in 5" shortcut: $85^2 = ?$

MEDIUM

A. 6425

B. 5625

C. 7220

D. 7225 (Correct)

Explanation: For a number $10a+5$ (here $a=8$), the square is $a \times (a+1)$ followed by 25: $8 \times 9 = 72$, so $85^2 = 7225$.

Q63. Calculate quickly using the $\times 11$ shortcut: $68 \times 11 = ?$

MEDIUM

A. 748 (Correct)

B. 79

C. 680

D. 816

Explanation: Sandwich the digit sum of 68 between its own digits (carrying if the sum reaches 10 or more): $68 \times 11 = 748$.

Q64. Calculate quickly using the $\times 11$ shortcut: $57 \times 11 = ?$

MEDIUM

A. 684

B. 627 (Correct)

C. 68

D. 570

Explanation: Sandwich the digit sum of 57 between its own digits (carrying if the sum reaches 10 or more): $57 \times 11 = 627$.

Q65. Calculate quickly using the $\times 11$ shortcut: $75 \times 11 = ?$

MEDIUM

A. 86

B. 750

C. 900

D. 825 (Correct)

Explanation: Sandwich the digit sum of 75 between its own digits (carrying if the sum reaches 10 or more): $75 \times 11 = 825$.

Q66. Calculate quickly: $65^2 = ?$

MEDIUM

A. 4025

B. 4125

C. 4225 (Correct)

D. 4325

Explanation: For a number ending in 5: $6 \times (6+1) = 42$, then append 25 '4225.

Q67. Use the all-from-9-last-from-10 method: $1000 - 587 = ?$

MEDIUM

A. 313

B. 413 (Correct)

C. 423

D. 513

Explanation: $9-5=4$, $9-8=1$, $10-7=3$ '413.

Q68. How many factors does 12 have (including 1 and 12 itself)?

MEDIUM

A. 5

B. 7

C. 6 (Correct)

D. 8

Explanation: The factors of 12 are $\{1, 2, 3, 4, 6, 12\}$ — that's 6 in total.

Q69. How many factors does 18 have (including 1 and 18 itself)?

MEDIUM

- A. 5
- B. 7
- C. 8
- D. 6 (Correct)**

Explanation: The factors of 18 are {1, 2, 3, 6, 9, 18} — that's 6 in total.

Q70. How many factors does 24 have (including 1 and 24 itself)?

MEDIUM

- A. 8 (Correct)**
- B. 7
- C. 9
- D. 10

Explanation: The factors of 24 are {1, 2, 3, 4, 6, 8, 12, 24} — that's 8 in total.

Q71. How many factors does 20 have (including 1 and 20 itself)?

MEDIUM

- A. 5
- B. 6 (Correct)**
- C. 7
- D. 8

Explanation: The factors of 20 are {1, 2, 4, 5, 10, 20} — that's 6 in total.

Q72. How many factors does 28 have (including 1 and 28 itself)?

MEDIUM

- A. 5
- B. 7
- C. 6 (Correct)**
- D. 8

Explanation: The factors of 28 are {1, 2, 4, 7, 14, 28} — that's 6 in total.

Q73. What is the LCM (Lowest Common Multiple) of 4 and 6?

MEDIUM

- A. 24
- B. 12 (Correct)**
- C. 16
- D. 6

Explanation: $\text{LCM}(4, 6) = (4 \times 6) \div \text{HCF}(4, 6) = (24) \div 2 = 12$.

Q74. What is the LCM (Lowest Common Multiple) of 8 and 12?

MEDIUM

- A. 96
- B. 32
- C. 24 (Correct)**
- D. 12

Explanation: $\text{LCM}(8, 12) = (8 \times 12) \div \text{HCF}(8, 12) = (96) \div 4 = 24$.

Q75. What is the LCM (Lowest Common Multiple) of 9 and 15?

MEDIUM

- A. 135
- B. 54
- C. 15
- D. 45 (Correct)**

Explanation: $\text{LCM}(9, 15) = (9 \times 15) \div \text{HCF}(9, 15) = (135) \div 3 = 45$.

Q76. What is the LCM (Lowest Common Multiple) of 6 and 10?

MEDIUM

- A. 30 (Correct)**
- B. 60
- C. 36
- D. 10

Explanation: $\text{LCM}(6, 10) = (6 \times 10) \div \text{HCF}(6, 10) = (60) \div 2 = 30$.

Q77. What is the LCM (Lowest Common Multiple) of 14 and 21?

MEDIUM

- A. 294
- B. 42 (Correct)**
- C. 56
- D. 21

Explanation: $\text{LCM}(14, 21) = (14 \times 21) \div \text{HCF}(14, 21) = (294) \div 7 = 42$.

Q78. Is 17 a Prime or a Composite number?

MEDIUM

- A. Composite
- B. Even
- C. A perfect square
- D. Prime (Correct)**

Explanation: 17 has no factors other than 1 and itself (checking all numbers up to 17 confirms none divide it evenly), so it is Prime.

Q79. Is 21 a Prime or a Composite number?

MEDIUM

- A. Composite (Correct)**
- B. Prime
- C. Even
- D. A perfect square

Explanation: $21 = 3 \times 7$ — it has factors beyond 1 and itself, so it is Composite.

Q80. Is 29 a Prime or a Composite number?

MEDIUM

- A. Composite
- B. Prime (Correct)**
- C. Even
- D. A perfect square

Explanation: 29 has no factors other than 1 and itself (checking all numbers up to 29 confirms none divide it evenly), so it is Prime.

Q81. Is 33 a Prime or a Composite number?

MEDIUM

- A. Prime
- B. Even
- C. Composite (Correct)**
- D. A perfect square

Explanation: $33 = 3 \times 11$ — it has factors beyond 1 and itself, so it is Composite.

Q82. Is 41 a Prime or a Composite number?

MEDIUM

- A. Composite
- B. Even
- C. A perfect square
- D. Prime (Correct)**

Explanation: 41 has no factors other than 1 and itself (checking all numbers up to 41 confirms none divide it evenly), so it is Prime.

Q83. Which of these fractions is the LARGEST: $\frac{2}{3}$, $\frac{1}{2}$, $\frac{3}{8}$, $\frac{1}{4}$?

MEDIUM

- A. $\frac{1}{2}$
- B. $\frac{3}{8}$
- C. $\frac{1}{4}$
- D. $\frac{2}{3}$ (Correct)**

Explanation: Converting each to a decimal: $\frac{2}{3}=0.67$, $\frac{1}{2}=0.5$, $\frac{3}{8}=0.38$, $\frac{1}{4}=0.25$. The largest is $\frac{2}{3}$.

Q84. Which of these fractions is the LARGEST: $\frac{5}{6}$, $\frac{2}{3}$, $\frac{1}{2}$, $\frac{3}{10}$?

MEDIUM

- A. $\frac{5}{6}$ (Correct)**
- B. $\frac{2}{3}$
- C. $\frac{1}{2}$
- D. $\frac{3}{10}$

Explanation: Converting each to a decimal: $\frac{5}{6}=0.83$, $\frac{2}{3}=0.67$, $\frac{1}{2}=0.5$, $\frac{3}{10}=0.3$. The largest is $\frac{5}{6}$.

Q85. Which of these fractions is the LARGEST: $\frac{3}{5}$, $\frac{7}{10}$, $\frac{2}{5}$, $\frac{1}{3}$?

MEDIUM

- A. $\frac{3}{5}$
- B. $\frac{7}{10}$ (Correct)**
- C. $\frac{2}{5}$
- D. $\frac{1}{3}$

Explanation: Converting each to a decimal: $\frac{3}{5}=0.6$, $\frac{7}{10}=0.7$, $\frac{2}{5}=0.4$, $\frac{1}{3}=0.33$. The largest is $\frac{7}{10}$.

Q86. Which of these fractions is the LARGEST: $\frac{4}{9}$, $\frac{5}{6}$, $\frac{1}{4}$, $\frac{2}{9}$?

MEDIUM

- A. $\frac{4}{9}$
- B. $\frac{1}{4}$
- C. $\frac{5}{6}$ (Correct)**
- D. $\frac{2}{9}$

Explanation: Converting each to a decimal: $\frac{4}{9}=0.44$, $\frac{5}{6}=0.83$, $\frac{1}{4}=0.25$, $\frac{2}{9}=0.22$. The largest is $\frac{5}{6}$.

Q87. Which of these fractions is the LARGEST: $\frac{5}{8}$, $\frac{3}{4}$, $\frac{1}{2}$, $\frac{3}{8}$?

MEDIUM

- A. $\frac{5}{8}$
- B. $\frac{1}{2}$
- C. $\frac{3}{8}$
- D. $\frac{3}{4}$ (Correct)**

Explanation: Converting each to a decimal: $\frac{5}{8}=0.63$, $\frac{3}{4}=0.75$, $\frac{1}{2}=0.5$, $\frac{3}{8}=0.38$. The largest is $\frac{3}{4}$.

Q88. Calculate: $\frac{2}{3} \times \frac{3}{4} = ?$

MEDIUM

- A. $\frac{17}{12}$
- B. $\frac{6}{13}$
- C. $\frac{1}{2}$ (Correct)**
- D. $\frac{9}{8}$

Explanation: Multiply numerators together and denominators together: $(2 \times 3)/(3 \times 4) = 6/12$, which simplifies to $\frac{1}{2}$.

Q89. Calculate quickly using the "all from 9, last from 10" shortcut: $10000 - 6789 = ?$

HARD

- A. 3221
- B. 3211 (Correct)**
- C. 3210
- D. 3212

Explanation: Subtract every digit of 6789 from 9 except the last, which is subtracted from 10: $10000 - 6789 = 3211$.

Q90. Calculate quickly using the "all from 9, last from 10" shortcut: $10000 - 4321 = ?$

HARD

- A. 5689
- B. 5679 (Correct)**
- C. 5678
- D. 5680

Explanation: Subtract every digit of 4321 from 9 except the last, which is subtracted from 10: $10000 - 4321 = 5679$.

Q91. Calculate quickly using the "all from 9, last from 10" shortcut: $10000 - 9876 = ?$

HARD

- A. 124 (Correct)**
- B. 123
- C. 125
- D. 134

Explanation: Subtract every digit of 9876 from 9 except the last, which is subtracted from 10: $10000 - 9876 = 124$.

Q92. Calculate quickly using vertically-and-crosswise multiplication: $89 \times 76 = ?$

HARD

- A. 6764 (Correct)**
- B. 6794
- C. 6774
- D. 6754

Explanation: Units: $9 \times 6 = 54$. Cross-multiply and add: $8 \times 6 + 9 \times 7 = 111$. Leading: $8 \times 7 = 56$. Combine with carries: $89 \times 76 = 6764$.

Q93. Calculate quickly using vertically-and-crosswise multiplication: $93 \times 87 = ?$

HARD

- A. 8081
- B. 8091 (Correct)**
- C. 8051
- D. 8101

Explanation: Units: $3 \times 7 = 21$. Cross-multiply and add: $9 \times 7 + 3 \times 8 = 87$. Leading: $9 \times 8 = 72$. Combine with carries: $93 \times 87 = 8091$.

Q94. Calculate quickly using vertically-and-crosswise multiplication: $98 \times 67 = ?$

HARD

- A. 6576
- B. 6556
- C. 6566 (Correct)**
- D. 6596

Explanation: Units: $8 \times 7 = 56$. Cross-multiply and add: $9 \times 7 + 8 \times 6 = 111$. Leading: $9 \times 6 = 54$. Combine with carries: $98 \times 67 = 6566$.

Q95. Calculate quickly using the Nikhilam near-base shortcut: $998 \times 997 = ?$

HARD

- A. 995000
- B. 1001006
- C. 995006 (Correct)**
- D. 994994

Explanation: Both numbers are close to 1000, with deviations -2 and -3. Cross-add one number and the other's deviation ($998 - 3 = 995$), multiply by the base, then add the product of the deviations ($-2 \times -3 = 6$): $1000 \times 995 + 6 = 995006$.

Q96. Calculate quickly using the Nikhilam near-base shortcut: $1003 \times 1008 = ?$

HARD

- A. 1011024 (Correct)**
- B. 1010976
- C. 1011000
- D. 995024

Explanation: Both numbers are close to 1000, with deviations +3 and +8. Cross-add one number and the other's deviation ($1003 + 8 = 1011$), multiply by the base, then add the product of the deviations ($3 \times 8 = 24$): $1000 \times 1011 + 24 = 1011024$.

Q97. Calculate quickly using the Nikhilam near-base shortcut: $994 \times 991 = ?$

HARD

- A. 985000
- B. 1003054
- C. 985054 (Correct)**
- D. 984946

Explanation: Both numbers are close to 1000, with deviations -6 and -9. Cross-add one number and the other's deviation ($994 - 9 = 985$), multiply by the base, then add the product of the deviations ($-6 \times -9 = 54$): $1000 \times 985 + 54 = 985054$.

Q98. Calculate quickly using the "square ending in 5" shortcut: $105^2 = ?$

HARD

- A. 10025
- B. 9025
- C. 11020
- D. 11025 (Correct)**

Explanation: For a number $10a+5$ (here $a=10$), the square is $a \times (a+1)$ followed by 25: $10 \times 11 = 110$, so $105^2 = 11025$.

Q99. Calculate quickly using the "square ending in 5" shortcut: $115^2 = ?$

HARD

- A. 12125
- B. 11025
- C. 13220
- D. 13225 (Correct)**

Explanation: For a number $10a+5$ (here $a=11$), the square is $a \times (a+1)$ followed by 25: $11 \times 12 = 132$, so $115^2 = 13225$.

Q100. Calculate quickly using the "square ending in 5" shortcut: $125^2 = ?$

HARD

- A. 14425
- B. 13225
- C. 15620
- D. 15625 (Correct)**

Explanation: For a number $10a+5$ (here $a=12$), the square is $a \times (a+1)$ followed by 25: $12 \times 13 = 156$, so $125^2 = 15625$.

Q101. Calculate quickly using the $\times 11$ shortcut: $234 \times 11 = ?$

HARD

- A. 2340
- B. 2808
- C. 2574 (Correct)**
- D. 245

Explanation: Sandwich the digit sum of 234 between its own digits (carrying if the sum reaches 10 or more): $234 \times 11 = 2574$.

Q102. Calculate quickly using the $\times 11$ shortcut: $456 \times 11 = ?$

HARD

- A. 5016 (Correct)**
- B. 467
- C. 4560
- D. 5472

Explanation: Sandwich the digit sum of 456 between its own digits (carrying if the sum reaches 10 or more): $456 \times 11 = 5016$.

Q103. Calculate quickly using the $\times 11$ shortcut: $123 \times 11 = ?$

HARD

- A. 134
- B. 1230
- C. 1476
- D. 1353 (Correct)**

Explanation: Sandwich the digit sum of 123 between its own digits (carrying if the sum reaches 10 or more): $123 \times 11 = 1353$.

Q104. Calculate quickly: $98 \times 97 = ?$

HARD

- A. 9406
- B. 9506 (Correct)**
- C. 9606
- D. 9516

Explanation: Base 100, deviations -2 and -3: $(98-3) \times 100 + (-2 \times -3) = 9500 + 6 = 9506$.

Q105. Calculate quickly: $102 \times 104 = ?$

HARD

- A. 10408
- B. 10508
- C. 10608 (Correct)**
- D. 10708

Explanation: Base 100, deviations +2 and +4: $(102+4) \times 100 + (2 \times 4) = 10600 + 8 = 10608$.

Q106. Three bells ring at intervals of 4, 6, and 8 minutes respectively. If they all ring together now, after how many minutes will they next ring together?

HARD

- A. 18
- B. 12
- C. 24 (Correct)**
- D. 16

Explanation: The bells ring together again after the LCM of their intervals: $\text{LCM}(4, 6, 8) = 24$ minutes.

Q107. Three bells ring at intervals of 5, 10, and 15 minutes respectively. If they all ring together now, after how many minutes will they next ring together?

HARD

- A. 10
- B. 40
- C. 15
- D. 30 (Correct)**

Explanation: The bells ring together again after the LCM of their intervals: $\text{LCM}(5, 10, 15) = 30$ minutes.

Q108. Three bells ring at intervals of 6, 9, and 12 minutes respectively. If they all ring together now, after how many minutes will they next ring together?

HARD

- A. 36 (Correct)**
- B. 27
- C. 18
- D. 24

Explanation: The bells ring together again after the LCM of their intervals: $\text{LCM}(6, 9, 12) = 36$ minutes.

Q109. Three bells ring at intervals of 3, 4, and 6 minutes respectively. If they all ring together now, after how many minutes will they next ring together?

HARD

- A. 13
- B. 12 (Correct)**
- C. 18
- D. 3

Explanation: The bells ring together again after the LCM of their intervals: $\text{LCM}(3, 4, 6) = 12$ minutes.

Q110. Three bells ring at intervals of 8, 12, and 16 minutes respectively. If they all ring together now, after how many minutes will they next ring together?

HARD

- A. 36
- B. 24
- C. 48 (Correct)**
- D. 32

Explanation: The bells ring together again after the LCM of their intervals: $\text{LCM}(8, 12, 16) = 48$ minutes.

Q111. What is the HCF of 12, 18, and 24?

HARD

A. 6 (Correct)

B. 8

C. 10

D. 1

Explanation: First find $\text{HCF}(12, 18) = 6$, then $\text{HCF}(6, 24) = 6$. So $\text{HCF}(12, 18, 24) = 6$.

Q112. What is the HCF of 16, 24, and 40?

HARD

A. 10

B. 8 (Correct)

C. 12

D. 2

Explanation: First find $\text{HCF}(16, 24) = 8$, then $\text{HCF}(8, 40) = 8$. So $\text{HCF}(16, 24, 40) = 8$.

Q113. What is the HCF of 20, 30, and 50?

HARD

A. 12

B. 14

C. 10 (Correct)

D. 4

Explanation: First find $\text{HCF}(20, 30) = 10$, then $\text{HCF}(10, 50) = 10$. So $\text{HCF}(20, 30, 50) = 10$.

Q114. What is the HCF of 14, 21, and 28?

HARD

A. 9

B. 11

C. 1

D. 7 (Correct)

Explanation: First find $\text{HCF}(14, 21) = 7$, then $\text{HCF}(7, 28) = 7$. So $\text{HCF}(14, 21, 28) = 7$.

Q115. What is the HCF of 18, 27, and 45?

HARD

A. 9 (Correct)

B. 11

C. 13

D. 3

Explanation: First find $\text{HCF}(18, 27) = 9$, then $\text{HCF}(9, 45) = 9$. So $\text{HCF}(18, 27, 45) = 9$.

Q116. What is the LCM of 2, 3, and 4?

HARD

A. 6

B. 12 (Correct)

C. 24

D. 16

Explanation: First find $\text{LCM}(2, 3) = 6$, then $\text{LCM}(6, 4) = 12$. So $\text{LCM}(2, 3, 4) = 12$.

Q117. What is the LCM of 3, 4, and 6?

HARD

- A. 72
- B. 18
- C. 12 (Correct)**
- D. 3

Explanation: First find $\text{LCM}(3, 4) = 12$, then $\text{LCM}(12, 6) = 12$. So $\text{LCM}(3, 4, 6) = 12$.

Q118. What is the LCM of 4, 5, and 10?

HARD

- A. 10
- B. 200
- C. 28
- D. 20 (Correct)**

Explanation: First find $\text{LCM}(4, 5) = 20$, then $\text{LCM}(20, 10) = 20$. So $\text{LCM}(4, 5, 10) = 20$.

Q119. What is the LCM of 2, 5, and 6?

HARD

- A. 30 (Correct)**
- B. 10
- C. 60
- D. 34

Explanation: First find $\text{LCM}(2, 5) = 10$, then $\text{LCM}(10, 6) = 30$. So $\text{LCM}(2, 5, 6) = 30$.

Q120. What is the LCM of 3, 5, and 9?

HARD

- A. 15
- B. 45 (Correct)**
- C. 135
- D. 51

Explanation: First find $\text{LCM}(3, 5) = 15$, then $\text{LCM}(15, 9) = 45$. So $\text{LCM}(3, 5, 9) = 45$.

Q121. Convert 0.75 to a fraction in its simplest form.

HARD

- A. 75/100
- B. 19/25
- C. 4/3
- D. 3/4 (Correct)**

Explanation: $0.75 = 75/100$, which simplifies (dividing both by their HCF) to $3/4$.

Q122. Convert 0.4 to a fraction in its simplest form.

HARD

- A. 2/5 (Correct)**
- B. 4/10
- C. 1/2
- D. 5/2

Explanation: $0.4 = 4/10$, which simplifies (dividing both by their HCF) to $2/5$.

Q123. Convert 0.36 to a fraction in its simplest form.

HARD

- A. 36/100
- B. 9/25 (Correct)**
- C. 37/100
- D. 25/9

Explanation: $0.36 = 36/100$, which simplifies (dividing both by their HCF) to $9/25$.

Q124. Convert 0.15 to a fraction in its simplest form.

HARD

- A. 15/100
- B. 4/25
- C. 3/20 (Correct)**
- D. 20/3

Explanation: $0.15 = 15/100$, which simplifies (dividing both by their HCF) to $3/20$.

Q125. Convert 0.6 to a fraction in its simplest form.

HARD

- A. 6/10
- B. 7/10
- C. 5/3
- D. 3/5 (Correct)**

Explanation: $0.6 = 6/10$, which simplifies (dividing both by their HCF) to $3/5$.

Q126. Calculate: $2/3 \div 4/9 = ?$

HARD

- A. 3/2 (Correct)**
- B. 8/27
- C. 27/8
- D. 2/3

Explanation: Dividing by a fraction means multiplying by its reciprocal: $2/3 \times 9/4 = 18/12$, which simplifies to $3/2$.

Language (67 questions)

Q1. Which word in the sentence is a noun?

"The teacher wrote on the board."

EASY

- A. teacher (Correct)**
- B. wrote
- C. on
- D. the

Explanation: A noun is a naming word for a person, place, or thing. 'Teacher' names a person, so it is the noun in this sentence.

Q2. Choose the correct word to complete the sentence: I saw ___ elephant at the zoo yesterday.

EASY

- A. a
- B. an (Correct)**
- C. the
- D. (no article needed)

Explanation: 'Elephant' begins with a vowel sound, so the indefinite article 'an' is used. Since this is the first time the elephant is mentioned, the indefinite article is required, not 'the'.

Q3. Choose the correct verb to complete the sentence: She ___ to school every day.

EASY

- A. go
- B. goes (Correct)**
- C. going
- D. gone

Explanation: In the simple present tense, third-person singular subjects like 'she' take a verb ending in -s/-es. 'Goes' correctly agrees with 'she'.

Q4. Choose the grammatically correct sentence.

EASY

- A. The dogs barks loudly.
- B. The dogs bark loudly. (Correct)**
- C. The dog bark loudly.
- D. The dog barking loudly.

Explanation: The plural subject 'dogs' requires the base form of the verb without -s in the simple present tense, so 'bark' is correct.

Q5. Choose the correct preposition: I was born ___ 2014.

EASY

- A. in (Correct)**
- B. on
- C. at
- D. by

Explanation: The preposition 'in' is used with years, months, and longer periods of time, so 'in 2014' is correct.

Q6. Choose the synonym of 'happy'.

EASY

- A. sad
- B. joyful (Correct)**
- C. angry
- D. tired

Explanation: 'Joyful' means feeling great pleasure, which matches the meaning of 'happy'.

Q7. Choose the antonym of 'brave'.

EASY

- A. courageous
- B. bold
- C. cowardly (Correct)**
- D. strong

Explanation: 'Cowardly' means lacking courage, which is the direct opposite of 'brave'.

Q8. Choose the correctly spelled word.

EASY

- A. recieve
- B. receive (Correct)**
- C. receeve
- D. receve

Explanation: The correct spelling follows the pattern 'e' before 'i' after the letter 'c', giving 'receive'.

Q9. Which punctuation mark should end this sentence?

"What is your name"

EASY

- A. Full stop (.)
- B. Question mark (?) (Correct)**
- C. Exclamation mark (!)
- D. Comma (,)

Explanation: This sentence directly asks something, so it must end with a question mark.

Q10. Which word in the sentence is a verb?

"The bird sings sweetly in the morning."

EASY

- A. bird
- B. sings (Correct)**
- C. sweetly
- D. morning

Explanation: A verb shows an action or state. 'Sings' shows the action the bird performs, making it the verb.

Q11. வெற்றிடத்தை நிரப்பவும்: "She ___ to school every day."

EASY

- A. go
- B. goes (Correct)**
- C. going
- D. gone

Explanation: 'She' என்பது மூன்றாம் நபர் ஒருமை என்பதால், எளிய நிகழ்காலச் சொல்லுக்கு '-s' சேர்க்கப்படுகிறது: 'goes'.

Q12. 'Mouse' இன் சரியான பன்மை வடிவத்தைத் தேர்ந்தெடுக்கவும்:

EASY

- A. Mouses
- B. Mice (Correct)**
- C. Mices
- D. Mouse

Explanation: 'Mouse' இன் ஒழுங்கற்ற பன்மை 'Mice' ஆகும், இது உயிரொலியை மாற்றுவதன் மூலம் உருவாகிறது, '-s' சேர்த்து அல்ல.

Q13. 'Mouse' இன் சரியான பன்மை வடிவத்தைத் தேர்ந்தெடுக்கவும்:

EASY

- A. Mouses
- B. Mice (Correct)**
- C. Mouse's
- D. Mices

Explanation: 'Mouse' இன் ஒழுங்கற்ற பன்மை 'Mice' ஆகும், '-s' சேர்த்து அல்ல. விருப்பம் (a) Mouses மற்றும் (d) Mices தவறாக ஒரு வழக்கமான பன்மை இறுதியைச் சேர்க்கின்றன, (c) Mouse's என்பது உடைமை வடிவம், பன்மை அல்ல.

Q14. 'Happy' இன் ஒத்த பொருள் கொண்ட சொல்லைத் தேர்ந்தெடுக்கவும்:

EASY

- A. Sad
- B. Joyful (Correct)**
- C. Angry
- D. Tired

Explanation: 'Joyful' என்றால் மிகுந்த மகிழ்ச்சி உணர்வது, 'Happy' போலவே. 'Sad' மற்றும் 'Angry' எதிரான அல்லது தொடர்பற்ற உணர்வுகள், 'Tired' குறைந்த ஆற்றலைக் குறிக்கிறது, மகிழ்ச்சியை அல்ல.

Q15. வெற்றிடத்தை நிரப்பவும்: "She ___ to school every day."

EASY

- A. go
- B. goes (Correct)**
- C. going
- D. gone

Explanation: 'She' என்ற எழுவாய் மூன்றாம் இடம் ஒருமை, மேலும் இந்த வாக்கியம் ஒரு பழக்கமான செயலை விவரிக்கிறது ('every day'), எனவே எளிய நிகழ்கால வினைச்சொல் -s/-es முடிவைப் பெற வேண்டும்: 'goes'. 'Go' என்பது I/you/we/they உடன் பயன்படுத்தப்படுகிறது, she உடன் அல்ல; 'going' மற்றும் 'gone' ஆகியவற்றிற்கு ஒரு துணை வினை (is going / has gone) தேவை, இங்கே தனியாக நிற்க முடியாது.

Q16. "Mouse" இன் சரியான பன்மை வடிவத்தைத் தேர்ந்தெடுக்கவும்:

EASY

- A. Mouses
- B. Mice (Correct)**
- C. Mouse
- D. Miced

Explanation: 'Mouse' இன் ஒழுங்கற்ற பன்மை உயிர் ஒலியை மாற்றுவதன் மூலம் உருவாகிறது: 'Mice', '-s' அல்லது '-es' சேர்த்து அல்ல. 'Mouses' மற்றும் 'Miced' ஆகியவை வழக்கமான விதியை தவறாகப் பயன்படுத்துகின்றன, 'Mouse' என்பது மாறாத ஒருமை வடிவம் மட்டுமே.

Q17. 'Mouse' இன் சரியான பன்மை வடிவத்தைத் தேர்ந்தெடுக்கவும்:

EASY

- A. Mouses
- B. Mice (Correct)**
- C. Mouse
- D. Mices

Explanation: 'Mouse' இன் ஒழுங்கற்ற பன்மை 'Mice' ஆகும், '-s' அல்லது '-es' சேர்த்து அல்ல.

Q18. சரியான எழுத்துப்பிழையற்ற வார்த்தையைத் தேர்ந்தெடுக்கவும்:

EASY

- A. Recieve
- B. Receive (Correct)**
- C. Receeve
- D. Receve

Explanation: சரியான எழுத்துப்பிழை 'Receive' ஆகும். ஆங்கிலத்தில் 'i before e, except after c' என்ற விதி உள்ளது — இந்த வார்த்தையில் உயிரெழுத்து ஜோடிக்கு முன் 'c' வருவதால், வரிசை மாறி 'ei' ஆகிறது, இதனால் 'Recieve' அல்ல, 'Receive' உருவாகிறது.

Q19. "Mouse" (சிறிய விலங்கு) இன் சரியான பன்மை வடிவத்தைத் தேர்ந்தெடுக்கவும்:

EASY

- A. Mouses
- B. Mice (Correct)**
- C. Mices
- D. Mouse

Explanation: "Mouse" இன் ஒழுங்கற்ற பன்மை "Mice" ஆகும், இது '-s' அல்லது '-es' சேர்ப்பதற்குப் பதிலாக உயிரெழுத்தை மாற்றி உருவாக்கப்படுகிறது. "Mouses" மற்றும் "Mices" தவறாக வழக்கமான '-s'/'-es' விதியைப் பயன்படுத்துகின்றன, "Mouse" என்பது மாற்றப்படாத ஒருமை வடிவம் மட்டுமே.

Q20. "Happy" என்ற சொல்லுக்கு இணையான அர்த்தமுள்ள சொல்லைத் தேர்ந்தெடுக்கவும்:

EASY

- A. Sad
- B. Joyful (Correct)**
- C. Angry
- D. Tired

Explanation: "Joyful" என்றால் மிகுந்த மகிழ்ச்சியை உணர்வது, இது "Happy" போன்றது. "Sad" மற்றும் "Angry" எதிர் உணர்வுகள், "Tired" குறைந்த ஆற்றலைக் குறிக்கிறது, உணர்வை அல்ல, எனவே இவற்றில் எதுவும் இணைச்சொல் அல்ல.

Q21. Choose the correct synonym of the word 'ANCIENT'.

EASY

- A. Modern
- B. Old (Correct)**
- C. New
- D. Fresh

Explanation: 'Ancient' means belonging to the distant past (very old). Thus, 'Old' is the correct synonym.

Q22. Read the sentence and identify the synonym of the word 'ANCIENT': 'The museum housed several ancient manuscripts from the Vedic era.'

EASY

- A. Modern
- B. Very Old (Correct)**
- C. Fragile
- D. Expensive

Explanation: 'Ancient' refers to belonging to the distant past or very old times. Thus, 'Very Old' is the correct synonym.

Q23. Choose the grammatically correct sentence.

MEDIUM

- A. While I was cooking, the phone rang. (Correct)**
- B. While I cooking, the phone rang.
- C. While I was cook, the phone rang.
- D. While I am cooking, the phone rang.

Explanation: An ongoing past action interrupted by a shorter past action uses past continuous ('was cooking') alongside simple past ('rang').

Q24. Choose the grammatically correct sentence.

MEDIUM

- A. Everest is highest mountain in the world.
- B. Everest is the highest mountain in the world. (Correct)**
- C. Everest is a highest mountain in the world.
- D. Everest is an highest mountain in the world.

Explanation: Superlative adjectives naming a unique, one-of-a-kind item always take the definite article 'the'.

Q25. Choose the grammatically correct sentence.

MEDIUM

- A. The furniture in this room are old.
- B. The furniture in this room is old. (Correct)**
- C. The furnitures in this room are old.
- D. The furniture in this room were old.

Explanation: 'Furniture' is an uncountable noun and always takes a singular verb, so 'is' is correct here.

Q26. Choose the correct preposition: I have lived in this town ___ ten years.

MEDIUM

- A. since
- B. for (Correct)**
- C. from
- D. at

Explanation: 'For' is used with a duration or length of time, such as 'ten years', so it correctly completes this sentence.

Q27. Choose the synonym of 'enormous'.

MEDIUM

- A. tiny
- B. huge (Correct)**
- C. narrow
- D. quiet

Explanation: 'Huge' means extremely large in size, which matches the meaning of 'enormous'.

Q28. Choose the antonym of 'ancient'.

MEDIUM

- A. old
- B. modern (Correct)**
- C. historic
- D. antique

Explanation: 'Modern' means belonging to the present time, which is the direct opposite of 'ancient'.

Q29. Choose the grammatically correct sentence.

MEDIUM

- A. She don't like mangoes.
- B. She doesn't like mangoes. (Correct)**
- C. She not like mangoes.
- D. She doesn't likes mangoes.

Explanation: Negative simple present sentences with third-person singular subjects use 'doesn't' plus the base form of the verb.

Q30. Choose the correctly punctuated sentence.

MEDIUM

- A. I bought apples oranges and bananas.
- B. I bought apples, oranges, and bananas. (Correct)**
- C. I bought, apples oranges and bananas.
- D. I bought apples oranges, and bananas.

Explanation: Items in a list are separated by commas, with a comma also placed before 'and' preceding the final item.

Q31. Choose the grammatically correct sentence.

MEDIUM

- A. He runs quick.
- B. He runs quickly. (Correct)**
- C. He is running quick.
- D. He runs quickest.

Explanation: A verb like 'runs' must be modified by an adverb, and 'quickly' is the correct adverb form of 'quick'.

Q32. Choose the grammatically correct sentence.

MEDIUM

- A. I have seen that movie yesterday.
- B. I saw that movie yesterday. (Correct)**
- C. I have saw that movie yesterday.
- D. I seen that movie yesterday.

Explanation: A specific past time marker like 'yesterday' requires the simple past tense, so 'saw' is correct.

Q33. Choose the correct word: Please ___ your seatbelt before the car starts.

MEDIUM

- A. where
- B. wear (Correct)**
- C. were
- D. we're

Explanation: 'Wear' is the verb meaning to put on or fasten something, which fits the instruction about the seatbelt.

Q34. Choose the grammatically correct sentence.

MEDIUM

- A. Everyone have finished their homework.
- B. Everyone has finished their homework. (Correct)**
- C. Everyone has finished they homework.
- D. Everyone has finish their homework.

Explanation: 'Everyone' is a singular indefinite pronoun and takes the singular verb 'has', with the possessive 'their' before the noun 'homework'.

Q35. Choose the correct preposition: She is good ___ solving puzzles.

MEDIUM

- A. in
- B. at (Correct)**
- C. on
- D. with

Explanation: 'Good at' is the fixed expression used to describe skill or ability at an activity like solving puzzles.

Q36. Which word in the sentence is a conjunction?

"I wanted to play, but it started raining."

MEDIUM

- A. wanted
- B. but (Correct)**
- C. started
- D. raining

Explanation: A conjunction joins two clauses. 'But' connects the clause about wanting to play with the clause about it raining.

Q37. Choose the grammatically correct sentence.

MEDIUM

- A. Yesterday, I go to the market and bought vegetables.
- B. Yesterday, I went to the market and bought vegetables. (Correct)**
- C. Yesterday, I went to the market and buy vegetables.
- D. Yesterday, I going to the market and bought vegetables.

Explanation: Both actions happened at the same past time marked by 'yesterday', so both verbs must be in the simple past tense: 'went' and 'bought'.

Q38. 'Child' இன் சரியான பன்மை வடிவத்தைத் தேர்ந்தெடுக்கவும்:

MEDIUM

- A. Childs
- B. Childes
- C. Children (Correct)**
- D. Childrens

Explanation: 'Child' இன் ஒழுங்கற்ற பன்மை 'Children' ஆகும், '-s' அல்லது '-es' சேர்த்து அல்ல.

Q39. 'Happy' க்கு மிக நெருக்கமான பொருள் கொண்ட சொல்லைத் தேர்ந்தெடுக்கவும்:

MEDIUM

- A. Sad
- B. Joyful (Correct)**
- C. Angry
- D. Tired

Explanation: 'Joyful' என்றால் மிகுந்த மகிழ்ச்சியை உணர்வது, இது 'Happy' போன்றதே. 'Sad', 'Angry' மற்றும் 'Tired' ஆகியவை தொடர்பற்ற அல்லது எதிர் உணர்வுகள்.

Q40. சரியான எழுத்துப்பிழையற்ற சொல்லைத் தேர்ந்தெடுக்கவும்:

MEDIUM

- A. Recieve
- B. Receive (Correct)**
- C. Recieve
- D. Receve

Explanation: சரியான எழுத்துப்பிழை 'c' க்குப் பிறகைத் தவிர i, e க்கு முன் வரும்' என்ற விதியைப் பின்பற்றுகிறது: R-E-C-E-I-V-E. மற்ற தேர்வுகளில் எழுத்துகள் தவறான இடத்தில் உள்ளன அல்லது விடுபட்டுள்ளன.

Q41. வெற்றிடத்தை நிரப்பவும்: "She ___ to school every day."

MEDIUM

- A. go
- B. goes (Correct)**
- C. going
- D. gone

Explanation: 'She' என்ற எழுவாய் மூன்றாம் நபர் ஒருமை ஆகும், மேலும் இந்த வாக்கியம் எளிய நிகழ்காலத்தில் ஒரு வழக்கமான செயலை விவரிக்கிறது, இதற்கு '-s/-es' வினைச்சொல் வடிவமான 'goes' தேவை. 'go' என்பது I/you/we/they உடன் பயன்படுத்தப்படுகிறது, 'going' க்கு ஒரு துணை வினை (is going) தேவை, 'gone' என்பது இறந்தகால பகுதி, எளிய நிகழ்கால வடிவம் அல்ல.

Q42. சரியான எழுத்துப்பிழையற்ற சொல்லைத் தேர்ந்தெடுக்கவும்:

MEDIUM

- A. Recieve
- B. Receive (Correct)**
- C. Recieve
- D. Receve

Explanation: சரியான எழுத்துப்பிழை 'Receive' ஆகும், இது 'c' க்குப் பிறகைத் தவிர i, e க்கு முன்' என்ற விதியைப் பின்பற்றுகிறது — சொல்லில் 'c' இருப்பதால், 'e', 'i' க்கு முன் வருகிறது. விருப்பம் (a) Recieve, ei/ie வரிசையை தலைகீழாக்குகிறது, (c) Recieve மற்றும் (d) Receve தவறான உயிரெழுத்து வடிவங்களைக் கொண்டுள்ளன.

Q43. "Brave" இன் எதிர்ச்சொல்லைத் தேர்ந்தெடுக்கவும்:

MEDIUM

- A. Courageous
- B. Cowardly (Correct)**
- C. Bold
- D. Fearless

Explanation: 'Brave' என்றால் தைரியம் கொண்டிருத்தல். இதன் எதிர்ச்சொல் 'Cowardly' (தைரியமின்மை, ஆபத்தை எதிர்கொள்ள பயப்படுதல்). 'Courageous', 'Bold', 'Fearless' ஆகியவை அனைத்தும் 'Brave' இன் ஒத்த சொற்களே, எதிர்ச்சொற்கள் அல்ல — இது மாணவர்கள் உண்மையான எதிர்ச்சொல்லுக்குப் பதிலாக தொடர்புடைய சொல்லைத் தேர்ந்தெடுக்கும் ஒரு பொதுவான பொறியாகும்.

Q44. சரியான எழுத்துப்பிழையற்ற சொல்லைத் தேர்ந்தெடுக்கவும்:

MEDIUM

- A. Recieve
- B. Receive (Correct)**
- C. Recieve
- D. Receve

Explanation: சரியான எழுத்துப்பிழை 'Receive' ஆகும். இது 'i before e, except after c' என்ற விதியைப் பின்பற்றுகிறது — சொல்லில் 'c' இருப்பதால், 'e', 'i'க்கு முன் வருகிறது: 'ceive'. 'Recieve' i மற்றும் e ஐ மாற்றி இந்த விதியை மீறுகிறது, 'Receve' மற்றும் 'Receve' ஆகியவை அதே சொல்லின் தவறான எழுத்துப்பிழை வடிவங்கள் மட்டுமே.

Q45. வெற்றிடத்தை நிரப்பவும்: "She ___ to school every day."

MEDIUM

- A. go
- B. goes (Correct)**
- C. going
- D. gone

Explanation: எழுவாய் 'She' மூன்றாம் நபர் ஒருமை, மேலும் இந்த வாக்கியம் நிகழ்காலத்தில் ஒரு வழக்கமான செயலை விவரிக்கிறது, எனவே வினைச்சொல்லுடன் '-s' சேர்க்கப்படுகிறது: 'goes'. 'Go' என்பது I/you/we/they உடன் பயன்படுத்தப்படும்; 'going'-க்கு 'is' போன்ற துணை வினை தேவை; 'gone' என்பது இறந்தகால பகுதி (past participle).

Q46. 'Brave' இன் எதிர்ச்சொல்லைத் தேர்ந்தெடுக்கவும்:

MEDIUM

- A. Bold
- B. Coward (Correct)**
- C. Strong
- D. Fearless

Explanation: 'Coward' என்றால் தைரியம் இல்லாதவர், இது 'Brave'-க்கு எதிரானது. 'Bold', 'Strong' மற்றும் 'Fearless' ஆகியவை அனைத்தும் 'Brave'-இன் ஒத்த சொற்கள், எதிர்ச்சொற்கள் அல்ல.

Q47. வெற்றிடத்தை நிரப்பவும்: "Yesterday, she ___ to the market."

MEDIUM

- A. go
- B. goes
- C. went (Correct)**
- D. going

Explanation: "Yesterday" எனிய இறந்த காலத்தைக் குறிக்கிறது, "go" இன் இறந்த கால வடிவம் "went" ஆகும். "Go" மற்றும் "goes" நிகழ்கால வடிவங்கள், "going"க்கு இலக்கணப்படி முழுமையாக இருக்க "was" போன்ற துணை வினைச்சொல் தேவை, எனவே இவற்றில் எதுவும் இங்கு பொருந்தாது.

Q48. சரியான எழுத்துப்பிழையற்ற சொல்லைத் தேர்ந்தெடுக்கவும்:

MEDIUM

- A. Recieve
- B. Receive (Correct)**
- C. Receve
- D. Receeve

Explanation: சரியான எழுத்துப்பிழை "Receive" ஆகும் — விதியை நினைவில் கொள்ளுங்கள் "c க்குப் பிறகு தவிர, i முதலில் e க்கு முன்". "Recieve" இல் c க்குப் பிறகு i மற்றும் e மாற்றப்பட்டுள்ளன, "Receve"/"Recevee" இல் எழுத்துக்கள் விடுபட்டுள்ளன அல்லது கூடுதலாக சேர்க்கப்பட்டுள்ளன.

Q49. What is the opposite (antonym) of the word 'BRAVE'?

MEDIUM

- A. Bold
- B. Cowardly (Correct)**
- C. Heroic
- D. Strong

Explanation: 'Brave' means showing courage. Its direct opposite is 'Cowardly' (lacking courage).

Q50. Select the correct article to fill in the blank: 'He returned after ____ hour of intensive study.'

MEDIUM

- A. a
- B. an (Correct)**
- C. the
- D. no article required

Explanation: The word 'hour' begins with a silent 'h' and produces a vowel sound (/aʊ/), so the indefinite article 'an' is required before it.

Q51. Choose the grammatically correct sentence.

HARD

- A. By the time the guests arrived, she had cooked the meal. (Correct)**
- B. By the time the guests arrived, she has cooked the meal.
- C. By the time the guests arrived, she cooked the meal.
- D. By the time the guests arrived, she have cooked the meal.

Explanation: When one past action finished before another past action, the earlier action uses the past perfect tense ('had cooked'), while the later one uses simple past ('arrived').

Q52. Choose the grammatically correct sentence.

HARD

- A. Neither the teacher nor the students was late.
- B. Neither the teacher nor the students were late. (Correct)**
- C. Neither the teacher nor the students is late.
- D. Neither the teacher nor the students be late.

Explanation: With 'neither...nor', the verb agrees with the subject nearer to it. Since 'students' (plural) is nearer, the plural past-tense verb 'were' is correct.

Q53. Choose the grammatically correct sentence.

HARD

- A. The honesty is the best policy.
- B. Honesty is the best policy. (Correct)**
- C. An honesty is the best policy.
- D. A honesty is the best policy.

Explanation: Abstract, uncountable nouns used in a general sense, like 'honesty' here, are used without any article.

Q54. Choose the correct preposition: The students are afraid ___ the dark.

HARD

A. of (Correct)

- B. from
- C. with
- D. about

Explanation: 'Afraid of' is the standard fixed expression used to describe fear of something, such as the dark.

Q55. Choose the synonym of 'diligent'.

HARD

A. lazy

B. hardworking (Correct)

- C. careless
- D. forgetful

Explanation: 'Diligent' means showing careful and persistent effort in work, which matches the meaning of 'hardworking'.

Q56. Choose the antonym of 'generous'.

HARD

A. kind

B. selfish (Correct)

- C. wealthy
- D. polite

Explanation: 'Selfish' means concerned only with one's own interest and unwilling to share, which is the direct opposite of 'generous'.

Q57. Choose the grammatically correct sentence.

HARD

A. Walking to school, the rain started falling on Ravi.

B. Walking to school, Ravi got caught in the rain. (Correct)

- C. Walking to school, the school bag got wet.
- D. Walking to school, it started to rain on Ravi's bag.

Explanation: The introductory phrase 'Walking to school' must be immediately followed by the person actually doing the walking, which is 'Ravi'.

Q58. Choose the correctly punctuated sentence.

HARD

A. The dog's bone was buried in the yard. (Correct)

- B. The dogs bone was buried in the yard.
- C. The dog's bones was buried in the yard.
- D. The dogs' bone were buried in the yard.

Explanation: An apostrophe followed by 's' shows singular possession, so "dog's bone" correctly shows the bone belonging to one dog, matching the singular verb 'was'.

Q59. Choose the grammatically correct sentence.

HARD

A. This book is mine. (Correct)

- B. This book is my.
- C. This book is me.
- D. This book is mines.

Explanation: 'Mine' is the possessive pronoun that can stand alone as a complement to show ownership, so it correctly completes this sentence.

Q60. Choose the grammatically correct sentence.

HARD

- A. By next year, I will complete my project.
- B. By next year, I will have completed my project. (Correct)**
- C. By next year, I will completing my project.
- D. By next year, I have completed my project.

Explanation: The future perfect tense ('will have completed') is used to show an action that will be finished before a specific point in the future, marked here by 'by next year'.

Q61. Choose the grammatically correct sentence.

HARD

- A. The basket of apples are on the table.
- B. The basket of apples is on the table. (Correct)**
- C. The baskets of apples is on the table.
- D. The basket of apple are on the table.

Explanation: The true subject of the sentence is the singular noun 'basket', not 'apples' inside the prepositional phrase, so the singular verb 'is' is correct.

Q62. Choose the correct word: ___ going to the park after lunch.

HARD

- A. Their
- B. There
- C. They're (Correct)**
- D. Thier

Explanation: 'They're' is the contraction of 'they are', which correctly fits before the verb phrase 'going to the park'.

Q63. Choose the grammatically correct sentence.

HARD

- A. She likes reading, swimming, and to dance.
- B. She likes reading, swimming, and dancing. (Correct)**
- C. She likes to read, swimming, and dancing.
- D. She likes read, swim, and dance.

Explanation: In a list, all items joined by 'and' should share the same grammatical form; using the gerund form ('-ing') for all three items keeps the list parallel.

Q64. Choose the correctly punctuated sentence for the following direct speech.

HARD

- A. Riya said, I am going home.
- B. Riya said, "I am going home." (Correct)**
- C. Riya said "I am going home".
- D. "Riya said, I am going home."

Explanation: Direct speech needs a comma after the reporting verb, quotation marks around the exact words spoken, and the closing period inside the final quotation mark.

Q65. Choose the correct reported (indirect) speech for: She said, "I am tired."

HARD

- A. She said that she is tired.
- B. She said that she was tired. (Correct)**
- C. She said that she were tired.
- D. She said that she has tired.

Explanation: When the reporting verb 'said' is in the past tense, the present tense in the direct speech ('am') shifts back to the past tense ('was') in reported speech.

Q66. Choose one word for: 'A person who looks at the positive side of things'.

HARD

- A. Pessimist
- B. Optimist (Correct)**
- C. Atheist
- D. Orphan

Explanation: An 'Optimist' is someone who holds hope and sees the positive side. A 'Pessimist' sees the negative side.

Q67. Choose the correct ANTONYM for the word 'GENEROUS'.

HARD

- A. Kind
- B. Stingy (Correct)**
- C. Noble
- D. Humble

Explanation: 'Generous' means willing to give and share unselfishly. Its exact opposite is 'Stingy' (unwilling to spend or give).

General Knowledge (41 questions)

Q1. Which animal is recognised as the National Animal of India?

EASY

- A. Lion
- B. Tiger (Correct)**
- C. Elephant
- D. Leopard

Explanation: The Bengal Tiger is India's National Animal, chosen for its strength and grace, and India runs 'Project Tiger' for its conservation.

Q2. What is the capital city of India?

EASY

- A. Mumbai
- B. New Delhi (Correct)**
- C. Kolkata
- D. Chennai

Explanation: New Delhi is the capital of India, seat of the Parliament and the central government.

Q3. Which is the largest planet in our Solar System?

EASY

- A. Saturn
- B. Earth
- C. Jupiter (Correct)**
- D. Neptune

Explanation: Jupiter is the largest planet in the Solar System by both size and mass.

Q4. Which bird is the National Bird of India?

EASY

- A. Parrot
- B. Peacock (Correct)**
- C. Sparrow
- D. Crow

Explanation: The Indian Peacock is the National Bird of India, chosen for its vibrant colours and cultural significance.

Q5. How many continents are there on Earth?

EASY

- A. 5
- B. 6
- C. 7 (Correct)**
- D. 8

Explanation: Earth is divided into seven continents: Asia, Africa, North America, South America, Antarctica, Europe, and Australia.

Q6. From which direction does the Sun rise every day?

EASY

- A. North
- B. South
- C. East (Correct)**
- D. West

Explanation: Due to Earth's rotation, the Sun appears to rise in the east and set in the west.

Q7. Which sport is popularly known as the 'Gentleman's Game'?

EASY

- A. Football
- B. Hockey
- C. Cricket (Correct)**
- D. Tennis

Explanation: Cricket has traditionally been nicknamed the 'Gentleman's Game' because of its emphasis on fair play and sportsmanship.

Q8. What is the capital city of France?

EASY

- A. London
- B. Paris (Correct)**
- C. Rome
- D. Berlin

Explanation: Paris is the capital city of France, home to landmarks like the Eiffel Tower.

Q9. Which flower is the National Flower of India?

EASY

- A. Rose
- B. Sunflower
- C. Lotus (Correct)**
- D. Marigold

Explanation: The Lotus is India's National Flower, symbolising purity and prosperity in Indian culture.

Q10. What is the largest organ of the human body?

EASY

- A. Liver
- B. Heart
- C. Skin (Correct)**
- D. Brain

Explanation: The skin is the largest organ of the human body, covering and protecting the entire body.

Q11. நமது சூரிய குடும்பத்தில் மிகப்பெரிய கிரகம் எது?

EASY

- A. பூமி
- B. வியாழன் (Correct)**
- C. செவ்வாய்
- D. வெள்ளி

Explanation: நிறை மற்றும் அளவு ஆகிய இரண்டிலும் வியாழன் சூரிய குடும்பத்தின் மிகப்பெரிய கிரகம் ஆகும்.

Q12. Which river has been declared the National River of India?

MEDIUM

- A. Yamuna
- B. Godavari
- C. Ganga (Correct)**
- D. Brahmaputra

Explanation: The Ganga (Ganges) was declared the National River of India in 2008 due to its cultural, historical, and ecological importance.

Q13. Who founded the Scout movement in 1907?

MEDIUM

- A. Henry Dunant
- B. Robert Baden-Powell (Correct)**
- C. Pierre de Coubertin
- D. Alfred Nobel

Explanation: Robert Baden-Powell, a British Army officer, founded the Scout movement in 1907 to build character and outdoor skills in young people.

Q14. How many players are there in a cricket team on the field during a match?

MEDIUM

- A. 9
- B. 10
- C. 11 (Correct)**
- D. 12

Explanation: A cricket team fields 11 players at a time during a match.

Q15. Which is traditionally regarded as the longest river in the world?

MEDIUM

- A. Amazon
- B. Nile (Correct)**
- C. Yangtze
- D. Mississippi

Explanation: The Nile River in Africa is traditionally taught as the longest river in the world.

Q16. What is the capital city of Japan?

MEDIUM

- A. Osaka
- B. Tokyo (Correct)**
- C. Kyoto
- D. Seoul

Explanation: Tokyo is the capital city of Japan.

Q17. Which natural source is the body's main way of producing Vitamin D?

MEDIUM

A. Sunlight (Correct)

- B. Drinking water
- C. Table salt
- D. Moonlight

Explanation: When skin is exposed to sunlight, the body naturally produces Vitamin D.

Q18. Who founded the International Red Cross movement?

MEDIUM

A. Henry Dunant (Correct)

- B. Florence Nightingale
- C. Louis Pasteur
- D. Alexander Fleming

Explanation: Henry Dunant, a Swiss businessman, founded the International Red Cross in 1863 after witnessing the suffering of wounded soldiers.

Q19. Which is the smallest continent by land area?

MEDIUM

- A. Europe
- B. Antarctica
- C. Australia (Correct)**
- D. South America

Explanation: Australia is the smallest of the seven continents by land area.

Q20. How many players does one team have on the field in a standard game of football (soccer)?

MEDIUM

- A. 9
- B. 10
- C. 11 (Correct)**
- D. 12

Explanation: A standard football (soccer) team fields 11 players, including the goalkeeper.

Q21. In which Indian city is the Taj Mahal located?

MEDIUM

- A. Delhi
- B. Agra (Correct)**
- C. Jaipur
- D. Lucknow

Explanation: The Taj Mahal is located in Agra, Uttar Pradesh, built by Mughal emperor Shah Jahan.

Q22. Who was the first Prime Minister of independent India?

MEDIUM

- A. Mahatma Gandhi
- B. Jawaharlal Nehru (Correct)**
- C. Sardar Vallabhbhai Patel
- D. Dr. Rajendra Prasad

Explanation: Jawaharlal Nehru became India's first Prime Minister after independence in 1947.

Q23. Which is the largest hot desert in the world?

MEDIUM

- A. Thar Desert
- B. Sahara Desert (Correct)**
- C. Gobi Desert
- D. Arabian Desert

Explanation: The Sahara Desert in Africa is the largest hot desert in the world.

Q24. Which gas do green plants absorb from the air to make their food during photosynthesis?

MEDIUM

- A. Oxygen
- B. Carbon dioxide (Correct)**
- C. Nitrogen
- D. Hydrogen

Explanation: Green plants absorb carbon dioxide from the air and use sunlight to convert it into food, releasing oxygen as a by-product.

Q25. After how many years are the Summer Olympic Games usually held?

MEDIUM

- A. 2 years
- B. 3 years
- C. 4 years (Correct)**
- D. 5 years

Explanation: The Summer Olympic Games are traditionally held every four years.

Q26. What is the capital city of Australia?

MEDIUM

- A. Sydney
- B. Melbourne
- C. Canberra (Correct)**
- D. Perth

Explanation: Canberra is the capital city of Australia, specifically built to serve as the nation's capital.

Q27. Which country has the longest written constitution in the world?

HARD

- A. United States
- B. India (Correct)**
- C. United Kingdom
- D. Canada

Explanation: India's Constitution is the longest written constitution of any sovereign country in the world, containing a large number of articles and schedules.

Q28. How many Fundamental Rights are currently guaranteed by the Indian Constitution?

HARD

- A. 5
- B. 6 (Correct)**
- C. 7
- D. 8

Explanation: The Indian Constitution currently guarantees 6 Fundamental Rights, after the Right to Property was removed from the list by the 44th Constitutional Amendment in 1978.

Q29. Who was the first President of India?

HARD

- A. Dr. Rajendra Prasad (Correct)**
- B. Dr. S. Radhakrishnan
- C. Zakir Husain
- D. V.V. Giri

Explanation: Dr. Rajendra Prasad served as the first President of India from 1950 to 1962.

Q30. Which is the tallest mountain peak in the world above sea level?

HARD

- A. K2
- B. Kangchenjunga
- C. Mount Everest (Correct)**
- D. Nanga Parbat

Explanation: Mount Everest, located on the border of Nepal and Tibet, is the tallest mountain peak above sea level.

Q31. In India's three-tier Panchayati Raj system, what is the middle (intermediate) level called?

HARD

- A. Gram Sabha
- B. Panchayat Samiti (Correct)**
- C. Zila Parishad
- D. Lok Sabha

Explanation: The three-tier Panchayati Raj system consists of the Gram Panchayat (village level), Panchayat Samiti (block/intermediate level), and Zila Parishad (district level); Panchayat Samiti is the middle tier.

Q32. Who discovered the ABO human blood group system?

HARD

- A. Karl Landsteiner (Correct)**
- B. Louis Pasteur
- C. Edward Jenner
- D. Gregor Mendel

Explanation: Austrian scientist Karl Landsteiner discovered the ABO blood group system in 1901, a finding for which he later won the Nobel Prize.

Q33. How many bones are there in an average adult human body?

HARD

- A. 186
- B. 206 (Correct)**
- C. 226
- D. 246

Explanation: An average adult human body has 206 bones (babies are born with more bones that fuse together as they grow).

Q34. What is the approximate speed of light in vacuum?

HARD

- A. 3 lakh km per second (Correct)**
- B. 3 thousand km per second
- C. 30 lakh km per second
- D. 3 lakh km per hour

Explanation: Light travels at approximately 3,00,000 kilometres (3 lakh km) per second in a vacuum.

Q35. Who was the first human to walk on the Moon?

HARD

- A. Yuri Gagarin
- B. Neil Armstrong (Correct)**
- C. Buzz Aldrin
- D. Michael Collins

Explanation: Neil Armstrong became the first human to walk on the Moon on 20 July 1969 during the Apollo 11 mission.

Q36. N.R. Narayana Murthy is best known as a co-founder of which major Indian IT company?

HARD

- A. Wipro
- B. Infosys (Correct)**
- C. TCS
- D. HCL

Explanation: N.R. Narayana Murthy is one of the co-founders of Infosys, established in 1981, one of India's leading IT services companies.

Q37. Who wrote India's National Anthem, 'Jana Gana Mana'?

HARD

- A. Bankim Chandra Chattopadhyay
- B. Rabindranath Tagore (Correct)**
- C. Sarojini Naidu
- D. Muhammad Iqbal

Explanation: Rabindranath Tagore wrote and composed 'Jana Gana Mana', which was adopted as India's National Anthem.

Q38. How many spokes does the Ashoka Chakra on the Indian national flag have?

HARD

- A. 20
- B. 22
- C. 24 (Correct)**
- D. 26

Explanation: The Ashoka Chakra at the centre of the Indian national flag has 24 spokes.

Q39. Which Indian state has the longest coastline?

HARD

- A. Tamil Nadu
- B. Gujarat (Correct)**
- C. Andhra Pradesh
- D. Maharashtra

Explanation: Gujarat has the longest coastline among Indian states, stretching along the Arabian Sea.

Q40. Which planet is popularly known as the 'Red Planet'?

HARD

- A. Venus
- B. Mars (Correct)**
- C. Mercury
- D. Saturn

Explanation: Mars is called the 'Red Planet' because of the reddish appearance caused by iron oxide (rust) on its surface.

Q41. What is the chemical symbol for the element Gold?

HARD

- A. Go
- B. Gd
- C. Au (Correct)**
- D. Ag

Explanation: The chemical symbol for Gold is 'Au', derived from its Latin name 'aurum'.